

Universidade do Minho
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Proposal of an Approach for Inter-Organizational Data Architecture: Data Spaces

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Proposal of an Approach for Inter-Organizational Data Architecture: Data Spaces

Master's Dissertation Thesis
Master in Engineering and Management of Information Systems

Work done under the supervision of
Professor António Amaro Costa Vieira (PhD)

January 2026

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University of Minho, Braga, January 2026

Rui Pedro Fernandes Coelho

"Para ser grande, sê inteiro: nada
Teu exagera ou exclui.
Sê todo em cada coisa. Põe quanto és
No mínimo que fazes.
Assim em cada lago a lua toda
Brilha, porque alta vive."
(Reis, Ricardo, 1933)

AGRADECIMENTOS

Como todas as boas histórias, esta também tem o seu fim. E que história bonita esta foi. Não foi apenas um curso, apenas uma universidade, apenas estudo, apenas pessoas. Foi uma fase decisiva na minha vida. Uma fase de desconforto, trabalho, mudança e aprendizagem, mas que, no final, apenas posso recordar com a maior das felicidades. Não foi também uma fase linear, mas sim um processo de vários anos, em paralelo com mudanças, desafios e responsabilidades para além do contexto académico, onde a única constante foram as pessoas que estiveram ao meu lado durante todo o processo.

Em primeiro lugar, aos meus pais e avós. Pelo apoio, estabilidade e confiança silenciosa. Nunca faltaram, nunca pressionaram, e estiveram sempre lá. Ao meu irmão, Miguel, e Rita, pelo exemplo e por serem uma referência constante, pelos conselhos, pela presença, pelo equilíbrio que sempre trouxeram, e que tanto ajudou nas decisões importantes. Mais recentemente, ao Vicente, que chegou no meio de tudo isto e que, mesmo sem o saber, ajudou simplesmente pela sua existência, pelo sorriso e pela capacidade de lembrar do que realmente importa.

À minha Guida, pelo crescimento conjunto, pelo apoio diário, pela paciência nos momentos de maior cansaço e pela presença constante em todos os períodos particularmente exigentes. Ter alguém que compreende não só as palavras, mas também os silêncios, o stress e as ausências fez toda a diferença, não só aqui, como em todos os pontos da minha vida.

Aos meus amigos, desde os mais antigos até àqueles que conheci no início desta jornada. Todos foram essenciais para tornar esta jornada memorável. Foram presença em todos os momentos, foram casa mesmo quando longe, nas aulas, nas noites, de tricórnio ao peito e em tantas memórias que ficam, e noutras tantas que ainda estão por vir.

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RESUMO

Proposta de Abordagem para uma Arquitetura de Dados Inter-Organizacional: Data Spaces

A crescente relevância dos dados nas organizações contemporâneas tem intensificado a necessidade de arquiteturas capazes de otimizar a sua gestão, governação e reutilização. Neste contexto, esta dissertação propõe uma abordagem baseada em Data Spaces, analisando de que forma este paradigma pode mitigar desafios relacionados com a subutilização da informação, a redundância de dados e a falta de integração entre domínios pertencentes a organizações distintas.

O trabalho centra-se na conceção de uma arquitetura de dados descentralizada e federada, sustentada por princípios de soberania dos dados, interoperabilidade e governação distribuída. São analisados os principais componentes arquiteturais e artefactos necessários à sua implementação, bem como os desafios técnicos e organizacionais associados à adoção de Data Spaces em contextos distribuídos e inter-organizacionais.

A investigação segue a Design Science Research Methodology (DSRM), orientando o desenvolvimento de um artefacto arquitetural que materializa os conceitos estudados. Como resultado, é proposto um modelo de arquitetura de referência para Data Spaces aplicável a contextos inter-organizacionais, validado através de uma prova de conceito. Esta é realizada num cenário controlado, com organizações fictícias que representam participantes autónomos de um Data Space inter-organizacional, demonstrando a viabilidade da partilha controlada de dados entre domínios, preservando a autonomia dos produtores e promovendo a reutilização da informação.

Os resultados indicam que a adoção de uma arquitetura baseada em Data Spaces pode contribuir para a redução de redundâncias, a melhoria da interoperabilidade entre sistemas heterogéneos e o reforço da governação de dados. A abordagem proposta oferece um enquadramento arquitetural para organizações que procuram evoluir para modelos de gestão de dados mais distribuídos, flexíveis e alinhados com os princípios da economia dos dados, fornecendo ainda uma base estruturada para futuras implementações e investigações em ambientes inter-organizacionais.

Palavras-chave: Arquitetura de Dados**,** Data Spaces, Gestão de Dados, Governação de Dados, Interoperabilidade.

ABSTRACT

Proposal of an Approach for Inter-Organizational Data Architecture: Data Spaces

The increasing relevance of data in contemporary organizations has intensified the need for architectures capable of optimizing data management, governance, and reuse. In this context, this dissertation proposes an approach based on Data Spaces, analyzing how this paradigm can mitigate challenges related to data underutilization, information redundancy, and the lack of integration across domains belonging to distinct organizations.

This work focuses on the design of a decentralized and federated data architecture grounded on principles of data sovereignty, interoperability, and distributed governance. The main architectural components and artifacts required for its implementation are analyzed, along with the technical and organizational challenges associated with adopting Data Spaces in distributed and inter-organizational contexts.

The research follows the Design Science Research Methodology (DSRM), guiding the development of an architectural artifact that materializes the studied concepts. As a result, a reference architecture model for Data Spaces applicable to inter-organizational contexts is proposed and validated through a proof of concept. This proof of concept is conducted in a controlled scenario, using fictitious organizations that represent autonomous participants in an inter-organizational Data Space, demonstrating the feasibility of controlled data sharing across domains while preserving producer autonomy and promoting data reuse.

The results indicate that adopting a Data Spaces-based architecture can contribute to reducing data redundancy, improving interoperability between heterogeneous systems, and strengthening data governance. The proposed approach provides an architectural framework for organizations seeking to evolve towards more distributed and flexible data management models aligned with the principles of the data economy, while also offering a structured foundation for future implementations and research in inter-organizational environments.

Keywords: Data Architecture, Data Spaces, Data Management, Data Governance, Interoperability.

TABLE OF CONTENTS

Copyright and Terms of Use for Third Party Work ii

Statement of Integrity iii

Agradecimentos v

Resumo vi

Abstract vii

Table of Contents viii

List of Abbreviations and Acronyms xi

List of Figures xii

List of Tables xiii

1. Introduction 1

1.1 Scope and Motivation 1

1.2 Goals and Objectives of the Research 2

1.3 Methodological Approach 3

1.4 Literature Review Process 5

1.5 Document Structure 6

2. Fundamental Technical and Theoretical Concepts 8

2.1 The need for complex Data Management Architectures 8

2.2 Centralized vs Decentralized Data Management Paradigms 10

2.3 Centralized Data Management Paradigms 11

2.3.1 Data Warehouse 12

2.3.2 Data Lake 15

2.3.3 Data Lakehouse 18

2.3.4 Comparison between paradigms 19

2.4 Decentralized Data Management Paradigms 21

2.4.1 Data Mesh 22

2.4.2 Data Spaces 23
2.4.3 Comparison between paradigms 24
2.5 In Depth Look of Data Spaces 25
2.5.1 Definition of Data Spaces 26
2.5.2 Data Spaces Environment 26
2.5.3 Core Principles of Data Spaces 27
2.5.4 The Need for Data Spaces: Why They Emerged 28
2.5.5 Key Components of a Data Space 29
2.5.6 Data Governance in Data Spaces 30
3. Proposal of an Approach for Inter-Organizational Data Architecture: Data Spaces 31
3.1 Data Spaces Domain Model 31
3.1.1 Overview and Guiding Principles 32
3.1.2 Core Actors and Roles 33
3.1.3 Information and Policy Layer 34
3.1.4 Federated Trust and Governance Framework 35
3.1.5 System Layer: Connectors and Sovereign Exchange 36
3.1.6 Organizational Context and Maturity 37
3.2 Data Space Architecture 38
3.2.1 Data Space Conceptual Architecture 38
3.2.2 Data Space Technological Architecture 47
4. Data Spaces Proof of Concept 53
4.1 Scope of the Proof of Concept 53
4.2 Proof Of Concept Design Principles 59
4.3 Proof of Concept Implementation 60
4.3.1 Participant Definition and Trust Establishment 60
4.3.2 Data Product Publication 61

4.3.3 Contract Definition and Data Sharing Agreements 62
4.3.4 Governance Workflow and Access Grants 63
4.3.5 Federated Catalog Registration 66
4.3.6 Analytics Data Product Implementation 67
4.3.7 Proof Of Concept Artifact Mapping 68
4.4 Validation of the Proposed Approach 70
5. Final conclusions and future work 72
5.1 Conclusions of the proposed approach 73
5.2 Limitations and Future work 74
References 77
Appendix 81

List of Abbreviations and Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
ACID	Atomicity, Consistency, Isolation, Durability
BI	Business Intelligence
BDW	Big Data Warehouse
CDO	Chief Data Officer
CI/CD	Continuous Integration / Continuous Deployment
CRM	Customer Relationship Management
DW	Data Warehouse
DL	Data Lake
DS	Data Space
DSSP	Data Space Support Platform
DSRM	Design Science Research Methodology
ELT	Extract, Load, Transform
ERP	Enterprise Resource Planning
ETL	Extract, Transform, Load
GDPR	General Data Protection Regulation
HDFS	Hadoop Distributed File System
IDSA	International Data Spaces Association
IDS-RAM	International Data Spaces Reference Architecture Model
IoT	Internet of Things
JSON	JavaScript Object Notation
ML	Machine Learning
NoSQL	NoSQL
OLAP	Online Analytical Processing
PII	Personally Identifiable Information
PoC	Proof of Concept
SLA	Service-Level Agreement

LIST OF FIGURES

Figure 1: Design Science Research Methodology for Information Systems. Adapted from (Peffers et al., 2007) 3
Figure 2: Data Spaces Domain Model 32
Figure 3: Data Spaces Conceptual Architecture 40
Figure 4: Data Spaces Technological Architecture 48
Figure 5: Data Space PoC Scope / Architecture 54
Figure 6: PoC Participants: *dataspace-registry\participants\participants.yml* 82
Figure 7: PoC Data Product Definition: *domain-crm\customer_profile\product.yml* 84
Figure 8: PoC Data Contract: *customer_profile\contracts\baseline_contract.yml* 85
Figure 9: PoC Data Contract: *operational_metrics\contracts\org-c-analytics-provider.yml* .. 87
Figure 10: PoC Access Grants: *dataspace-registry\governance\access_grants.yml* 88
Figure 11: PoC Access Request – GitHub Issue 65
Figure 12: PoC Federated Catalog: dataspace-registry\catalog\catalog.json 90
Figure 13: PoC Artifact Mapping 69

LIST OF TABLES

Table 1: Comparison Between Data Management Paradigms 19
Table 2: Comparison Between Data Mesh and Data Spaces Paradigms 24
Table 3: Schema of Customer Profile Data Product (Organization A) 56
Table 4: Schema of Operational Metrics Data Product (Organization B) 57
Table 5: Schema of Engagement Insights Data Product (Organization C) 58

1. INTRODUCTION

This chapter presents the scope and motivation for this master's thesis, elaborating on the identified problem that justified its development. It details the purpose and objectives of the research, defines the methodological approach adopted, and describes the literature review process that was used. In the end, it provides an overview of the document's structure, outlining how the research is organized and presented.

1.1 Scope and Motivation

In modern organizations, data has become a fundamental asset, driving decision-making, operational efficiency, and strategic growth (Davenport, 2006; Manyika, 2011). However, as data volume, variety, and velocity continue to increase, traditional centralized data management architectures face growing limitations (Gandomi & Haider, 2015). Solutions such as Data Warehouses and Data Lakes, while widely implemented, struggle to support decentralized data operations, cross-organizational interoperability, and dynamic governance requirements (Inmon, 2005; Miloslavskaya & Tolstoy, 2016). As data ecosystems expand beyond individual enterprises, the need for federated data-sharing frameworks has become more urgent (Franklin et al., 2005).

The concept of Data Spaces has emerged as a response to these challenges, offering a structured and decentralized approach to data management that prioritizes data sovereignty, security, and interoperability (International Data Spaces Association [IDSA], 2019). Unlike traditional architectures that consolidate data into a single system, Data Spaces enable secure data exchange across multiple stakeholders while allowing data providers to retain control over access and usage conditions (Otto et al., 2019). This federated model ensures compliance with industry regulations and data governance policies, making it particularly relevant in environments where data privacy, trust, and transparency are critical factors (Otto, 2011).

By defining standards, governance models, and interoperability mechanisms, Data Spaces provide organizations with a scalable and trust-driven framework to facilitate secure and efficient data collaboration (IDSA, 2019). The adoption of this model aligns with recent regulatory initiatives, such as the European Data Strategy, which emphasizes the importance of sovereign and standardized data-sharing ecosystems across industries (European Commission, 2023). Given the increasing reliance on data-driven insights and cross-industry collaboration,

the implementation of Data Spaces represents a significant step toward more transparent, efficient, and regulatory-compliant data management practices.

This dissertation aims to explore the practical implementation of Data Spaces, investigating how they can optimize data governance, improve interoperability, and enhance organizational data quality. It examines their role in fostering trusted data-sharing environments, evaluates the technical and governance challenges involved, and assesses their impact on decision-making and operational efficiency. Through this research, the study will contribute to the growing discourse on Data Spaces, offering insights into their potential applications, advantages, and limitations.

1.2 Goals and Objectives of the Research

The primary objective of this dissertation is to design and evaluate a reference architecture for Data Spaces, assessing its effectiveness in supporting inter-organizational data management, improving data availability and quality, and enabling trusted, governed, and interoperable data sharing across autonomous organizations.

Given the limitations of traditional data architectures, such as Data Warehouses, Data Lakes, and even Data Mesh, this research aims to define a structured and federated framework that enables efficient, scalable, and compliant data exchange across multiple organizations.

To achieve this goal, the study will focus on the following key research objectives:

1. Define a reference architecture for Data Spaces, identifying the key components, governance mechanisms, and interoperability requirements necessary for their effective implementation. The architecture will align with existing frameworks, such as the International Data Spaces Reference Architecture Model (IDS-RAM) proposed by the International Data Spaces Association (IDSA, 2019), while addressing inter-organizational data-sharing challenges.
2. Investigate how Data Products and Data Contracts can be integrated into Data Spaces, ensuring that data governance policies, access control mechanisms, and interoperability standards are effectively enforced. This includes analyzing how Data Contracts regulate data-sharing agreements while maintaining compliance with legal and industry-specific requirements.
3. Evaluate the impact of Data Spaces on inter-organizational data management and decision-making, analyzing how they improve data accessibility, trust, security, and

interoperability while addressing inefficiencies in cross-enterprise collaboration. This objective will examine existing literature and practical implementations to understand the benefits and challenges associated with adopting Data Spaces.

4. Develop a Proof of Concept (PoC) to validate the proposed framework, demonstrating its feasibility in a controlled inter-organizational scenario based on fictitious organizations. The PoC will serve as a practical validation of the reference architecture, testing its technical capabilities, governance structures, interoperability mechanisms, and compliance-oriented artifacts.

By addressing these objectives, this dissertation aims to contribute to the evolution of inter-organizational data-sharing ecosystems, providing a structured and scalable model that organizations can adopt to facilitate trusted, governed, and efficient data collaboration.

1.3 Methodological Approach

To ensure rigor and reliability in the results obtained in this research, the Design Science Research Methodology (DSRM) for Information Systems was adopted. This methodology consists of six structured activities, designed to effectively and efficiently address problems in the field of Information Systems research (Peffers et al., 2007).

As proposed by Peffers et al., (2007), the methodology consists of six interconnected activities that generally follow a sequential order, progressing from the first to the sixth activity. However, it does not impose a strictly linear process. Instead, it provides flexibility in execution, allowing the research to begin at any stage depending on the context, and enable a return to previous activities whenever necessary (Peffers et al., 2007). Figure 1 illustrates the methodology applied in the present master's thesis.

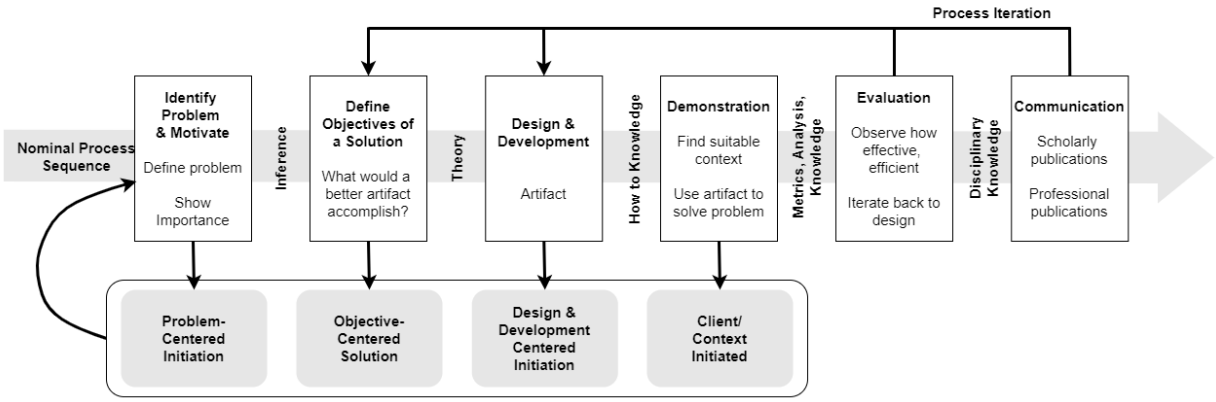


Figure 1: Design Science Research Methodology for Information Systems. Adapted from (Peffers et al., 2007)

As described above, the methodology consists of iterative and interconnected phases, providing the necessary flexibility for adjustments throughout the process, ultimately aiming at the development and presentation of a real-world solution.

- Activity 1 - Identify Problem and Motivate: This phase focuses on clearly defining the problem of inefficient management, underutilization, and limited interoperability of data across organizational boundaries. Additionally, it is essential to justify the relevance of the problem and the need for a viable solution, based on literature findings and the recognized potential of Data Spaces to support secure, decentralized, and policy-driven data sharing. Establishing the practical and theoretical motivation ensures a strong foundation for the research.
- Activity 2 - Define Objectives of a Solution: At this stage, the focus shifts to determining the specific objectives of the proposed solution. These objectives should be directly derived from the problem analysis, setting achievable and feasible goals while conducting the literature review on data management architectures, defining a reference architecture for Data Spaces, and developing a proof of concept to validate the proposed architecture in practice. Ensuring clarity in the definition of objectives, whether quantitative or qualitative, is crucial for guiding the subsequent development process.
- Activity 3 - Design and Development: This phase involves the conceptualization and development of the artifact, which in this case consists of designing the reference architecture for Data Spaces and developing its components, as described in the reference architecture. The phase also includes selecting appropriate technologies and tools and culminates with the development of a functional proof of concept to practically instantiate the proposed solution.
- Activity 4 - Demonstration: The fourth phase focuses on demonstrating the applicability of the developed solution through the implemented proof of concept, in solving specific aspects of the problem. The demonstration validates how the Data Spaces architecture facilitates improved data sharing, interoperability, governance, and decision-making within a controlled inter-organizational scenario. The emphasis is on providing concrete evidence that supports the feasibility of the proposed approach.

- Activity 5 - Evaluation: In this stage, the artifact is assessed by comparing the achieved results with the initially defined objectives. It consists of evaluating the Data Spaces reference architecture and PoC, considering aspects such as architectural alignment, governance coherence, interoperability, traceability, data quality, and usability. This process determines the effectiveness and suitability of the proposed solution, highlighting any necessary refinements or improvements.
- Activity 6 - Communication: The final phase encompasses the structured documentation and dissemination of the research outcomes, highlighting the innovation, utility, and contributions to both academic knowledge and practical needs. The communication occurs through the dissertation report, presentations, and discussions with interested parties in this matter.

1.4 Literature Review Process

To ensure a comprehensive and effective literature review, a structured process was established to identify and analyze relevant sources that form the foundation of this dissertation. The goal was to build a solid understanding of the current state of the art, while also identifying knowledge gaps that this research aims to address.

Given the novelty of Data Spaces and the limited availability of scientific literature on the subject, the review extended beyond traditional academic articles to include other reliable resources such as conference proceedings, white papers, webinars, or technical blogs. Only sources deemed objective and free from commercial bias were considered, ensuring the credibility of the information used.

The focus of the review was primarily on recently published content to capture the latest advancements in the field. Exceptions were made for foundational concepts like Data Warehouses, Data Lakes, and Big Data Warehouses, where earlier research remains highly relevant and influential.

A variety of keywords, including “Data Spaces”, “Data Products”, “Data Contracts”, “Data Lake”, “Big Data”, “Data Mesh”, “Data Management”, and other concepts were used to search academic databases such as Scopus, Google Scholar, ResearchGate, and Mendeley to ensure a broad and relevant selection of sources.

The selection process started with an initial screening of abstracts or introductions to determine relevance, followed by a detailed review of the most pertinent materials. This approach ensured

a focused and efficient examination of existing research, allowing for the identification of key insights. Additionally, attention was given for ongoing updates in the field, enabling the inclusion of newly published materials as the research progresses.
By synthesizing information from these varied sources, the literature review provides a strong theoretical foundation for the proposed Data Spaces architecture, identifying challenges in traditional data management and supporting the development of innovative solutions.

1.5 Document Structure

This dissertation report is structured into five chapters, each addressing a specific aspect of the research.

The first chapter – Introduction – Provides the scope and motivation for the dissertation, presenting the identified research problem and detailing the objectives, research questions, and the methodology used in this study. It also outlines the structure of the dissertation and the overall approach taken to address the research problem.

In the second chapter – Fundamental Technical and Theoretical Concepts – Are introduced the essential technical and theoretical concepts that underpin the research. It explores concepts such as Big Data, Data Warehouses, Data Lakes, Data Spaces, Governance, and others, explaining how these concepts form the basis of the proposed architecture. Additionally, it addresses the limitations of traditional approaches such as Data Warehouses and Data Lakes.

Third chapter – Proposal of an Approach for Inter-Organizational Data Architecture: Data Spaces – Presents the proposed approach for designing an Inter-Organizational Data Architecture based on Data Space principles. It introduces the domain model, conceptual architecture, and technological architecture that together define how a Data Space can be structured within an organization. The chapter details the rationale behind the selected components, explains how they address current challenges in inter-organizational data management, and establishes the foundations required for the subsequent Proof of Concept.

Fourth chapter – Data Space Proof of Concept – Describes the Proof of Concept developed to demonstrate the feasibility of the proposed Inter-Organizational Data Space approach. It defines the scope of the PoC, presents the proposed architecture, and details the implemented

components, including domains, data products, governance mechanisms, and data contracts. Through this demonstration, the chapter illustrates how the Data Spaces architecture operates in practice and evaluates its effectiveness in supporting controlled, interoperable, and sovereign data sharing.

Fifth – Final Conclusions and Future Work – Synthesizes the main findings of the dissertation, reflecting on the insights gained from the literature review, the proposed architectural approach, and the Proof of Concept implementation. It highlights the contributions of the research to the field of Data Spaces, acknowledges existing limitations, and identifies relevant directions for future work. A final overview of consulted scientific and technical sources ensures academic rigor and transparency.

Lastly – References – Provides a comprehensive summary of all consulted scientific literature, technical documents, technical blogs, and other relevant information sources used throughout this dissertation. This chapter ensures that proper credit is given to the authors and researchers whose work contributed to the foundation of this study, maintaining academic integrity and transparency in research.

2. FUNDAMENTAL TECHNICAL AND THEORETICAL CONCEPTS

This chapter establishes the foundational knowledge necessary to understand the scope of this research and the concepts that underpin the proposed solution. By addressing both technical and theoretical aspects, it lays the groundwork for the understanding, design, and validation of Data Spaces architecture.

The chapter begins with a brief introduction to Data Spaces, setting the stage for a more in-depth discussion later. It then explores the distinction between centralized and decentralized data management approaches, emphasizing their respective advantages and limitations. This comparison provides context for understanding the main differences and advantages and disadvantages of each approach.

Following this, the discussion extends to existing data architectures, analyzing their roles, structures, and use cases, providing a comparative perspective to highlight where each paradigm excels or falls short.

With this foundation in place, the chapter delves deeper into Data Spaces, presenting their definition, the environment in which they operate, their core principles, the reasons for their emergence, their key components and governance mechanisms.

By synthesizing these concepts, this chapter justifies the necessity of innovative solutions like Data Spaces in addressing the challenges of managing large, diverse, and distributed datasets in modern organizations.

2.1 The need for complex Data Management Architectures

It is widely recognized by modern organizations that the role of Big Data, Analytics, and Data Management is critical in driving the organization's strategic objectives (Davenport, 2006;Manyika, 2011). However, there is still no clear consensus on the optimal way to structure data management and analytics efforts within an organization or on the core data processes that must be supported. As organizations strive to leverage data for business intelligence, decision-making, and competitive advantage, it becomes essential to establish scalable, interoperable, and well-governed data architectures that align with their strategic goals (Grossman & Siegel, 2014;Santos & Costa, 2020).

The architectures designed for handling Big Data were created to accommodate the increasing scale and complexity of data, that following Laney (2001) 3V's model – Volume, Variety and Velocity – which served as the basis for defining Big Data for over a decade (Santos & Costa, 2020), These three characteristics were considered the primary challenges in dealing with Big Data, shaping its fundamental attributes. However, continuous research in Big Data management revealed that the 3V's model was incomplete, leading to the addition of two more V's: Veracity and Value (Machado et al., 2022b).

Summarizing the meaning of each one of the proposed V's, authors refer to Volume as the ever-growing amount of data generated from the multiple sources and interactions with the services that organizations use daily (Costa & Santos, 2017). Additionally, the term "big" in Big Data is dynamic and constantly evolving, as storage and processing capacities continue to expand over time (Gandomi & Haider, 2015). As a result, Big Data is not classified by the size of the data, but rather by the continuous and exponential growth of generated data (Krishnan, 2013).

Variety represents the diversity of data formats and its structural heterogeneity, which can be classified in three distinct categories such as structured, like relational databases, semi-structured, composed from JSON files or XML logs, or unstructured, such as images, videos or social media posts (Santos & Costa, 2020). Handling this variety requires flexible and scalable data processing architectures (Gandomi & Haider, 2015).

The third V in the original model from Laney concerns Velocity, which describes, from one perspective the speed at which data is generated (Gandomi & Haider, 2015), or the velocity at which data must be processed and analyzed to meet real-time decision-making needs (Santos & Costa, 2020).

Looking at the last two V's added to the initial proposal, Veracity addresses the inevitable uncertainty, inconsistency, and potential inaccuracies within data. Data may come from unreliable sources, contain biases, or require validation or the need to be treated with the appropriate techniques and technologies (Gandomi & Haider, 2015) to ensure its quality and trustworthiness for meaningful insights. The fifth V, Value, represents the usefulness of data once it is processed and analyzed. Raw data often holds little intrinsic value, until it undergoes transformation through analytics, machine learning, or business intelligence techniques to generate actionable insights that drive decision-making (Gandomi & Haider, 2015).

Given all mentioned characteristics of Big Data identified in the literature, its definition may vary accordingly to the authors and context, with no single, universally accepted definition serving as the absolute reference. However, its core attributes are well established and widely recognized. This concept has been instrumental in highlighting the need for advanced data

management and storage technologies, as well as specialized architectural paradigms to handle the increasing complexity of data.

In the following sections, data management paradigms are categorized into Centralized and Decentralized approaches, each encompassing distinct methodologies and technologies. Within these categories, a deeper exploration of specific data management models will be presented, providing a comprehensive understanding of their structure, functionality, and role in modern data ecosystems.

2.2 Centralized vs Decentralized Data Management Paradigms

As previously analyzed, organizational data management is a fundamental component in driving an organization's strategic objectives (Davenport, 2006; Manyika, 2011). Regardless of an organization's size or industry, the choice between centralized and decentralized data management systems plays a crucial role in shaping business operations, data governance, and decision-making processes. Each approach presents distinct advantages and challenges, influencing scalability, security, accessibility, and compliance (Grossman & Siegel, 2014).

Centralized and decentralized data management paradigms differ not only in data storage, processing and access but also in how decision rights, governance policies, and data responsibilities are assigned. Otto, (2011) emphasizes that organizations must carefully design their data governance structures to align with their business goals, whether they choose a top-down centralized approach or a distributed decentralized model.

His findings suggest that no single model fits all organizations. Some companies may initially adopt a centralized model and later transition to a decentralized approach as their data needs evolve. Others may implement hybrid strategies, leveraging centralized governance with decentralized operational autonomy, ensuring both compliance and agility. Large enterprises must evaluate whether data centralization improves operational efficiency or if a distributed approach enhances their scalability and flexibility (Otto, 2011).

The decision between centralized and decentralized data management is not binary but rather a strategic choice that depends on an organization’s structure, industry, and regulatory requirements. Understanding the trade-offs between both paradigms allows organizations to align their data strategies with long-term business goals and technological advancements.

In analyzing these two approaches, several key aspects will be considered. First, the governance structure will be examined, focusing on how decision-making power, access rights, and compliance policies are distributed (Velu et al., 2013). Then, the scalability and operational

efficiency of each model will be evaluated, considering how data growth impacts performance, flexibility, and adaptability to evolving business needs. Another crucial aspect is data integration and interoperability, which determines how well different data sources communicate within and across organizations. Finally, the security and compliance dimension will be explored, assessing how centralized and decentralized approaches handle regulatory requirements, risk mitigation, and access control (Rülicke et al., 2024).

2.3 Centralized Data Management Paradigms

A centralized data management paradigm consolidates all data-related operations under a single governing entity, ensuring consistency, compliance, and standardized policies across the organization. In this model, data is collected, processed, and stored within a central repository, where governance mechanisms oversee data quality, security, and integrity. The main advantage of centralization lies in its ability to provide a Single Source of Truth, ensuring uniformity in reporting, analytics, and decision-making processes (Otto, 2011).

Governance:

In a centralized model, decision-making power is concentrated within a single governing entity, often managed by a central data governance team or a Chief Data Officer (CDO). This ensures that data policies, security protocols, and compliance measures are applied consistently across the organization. Access rights are tightly controlled, typically enforced through Identity and Access Management (IAM) frameworks, ensuring that only authorized personnel can retrieve, modify, or analyze specific datasets. Compliance policies are also standardized at the central level, ensuring that all departments adhere to regulatory requirements such as General Data Protection Regulation (GDPR), reducing the risk of non-compliance (Otto, 2011).

Scalability and Operational Efficiency

While centralized data management ensures strong governance and compliance, it can introduce scalability challenges as data volumes grow. A single repository can become a bottleneck, affecting performance and response times, particularly in organizations processing large-scale, real-time data (Machado et al., 2022a). Operational efficiency of centralized architectures is also constrained by rigid infrastructure and high processing demands, requiring expensive scaling solutions, such as vertical scaling or cloud-based central storage systems to handle increasing workloads.

Data integration and Interoperability

Centralized models excel in data integration, as all information follows a predefined schema, ensuring consistency and accuracy across all business applications. This makes it easier to transform, manage, and analyze data in a structured manner (Otto, 2011). However, integration ability between different systems can be restricted, as all integrations must conform to a centralized data schema, making it difficult to incorporate heterogeneous, unstructured, or external data sources.

Security

Organizations adopting centralized data management benefit from strong regulatory oversight, as security mechanisms and compliance policies are applied uniformly across all business units. IAM frameworks, data encryption, and access monitoring ensure that data remains protected from unauthorized access (Grossman & Siegel, 2014). However, the centralization of security creates a single point of failure, making the system vulnerable to cyberattacks or infrastructure outages that could disrupt access to critical data.

Despite these challenges, centralized data models remain widely used in multiple industries, either due to strict compliance and data integrity requirements, or simply because they are generally more simple to manage. Their structured approach to data security, governance, and reporting ensures that organizations maintain high levels of accuracy, reliability, and control over their data assets (Grossman & Siegel, 2014).

2.3.1 Data Warehouse

As previously explored, the exponential growth of data generated across all sectors of an organization has driven the need for a robust Data Management architecture or paradigm – one that not only stores organizational data efficiently but also enables the extraction of valuable insights through analytical processes. Data Warehouses emerged as one of the first complete solutions for this need and can be defined as a system designed to collect, store, and process operational data to generate analytical insights for decision-making. Kimball & Ross, (2013) describe a Data Warehouse as a system that ingests raw operational data and transforms it into structured analytical outputs, ensuring data consistency and accuracy. In contrast, Inmon, (2005) presents the Data Warehouse as a subject-oriented, integrated, non-volatile repository that evolves over time.

Golfarelli & Rizzi, (2009) further conceptualize Data Warehouses as a collection of techniques, methods, and tools that allow managers and analysts to perform strategic data analysis. While their definition remains intentionally broad, it reinforces the idea that Data Warehouses are not just databases but complete analytical ecosystems.

An effective Data Warehouse is defined by several fundamental characteristics that ensure its efficiency and reliability. Kimball & Ross, (2013) identified these characteristics (firstly mentioned in 1996) as essential for optimizing analytical processes and decision-making. One of the key attributes is accessibility, meaning that data should be easily retrievable for analytical purposes. Additionally, integration is crucial, requiring data from multiple sources to be harmonized into a unified schema to ensure consistency across the organization. Query flexibility allows users to analyze data from multiple perspectives, enabling deeper insights into business operations. Another critical characteristic is information consistency, ensuring that reports and queries yield reliable and uniform results, minimizing discrepancies in business intelligence outputs. Multidimensional representation is also a defining factor, as data should be structured in a way that supports OLAP (Online Analytical Processing) queries, allowing for efficient aggregation and drill-down analysis. Lastly, correctness and completeness are essential, meaning that stored data must be accurate, validated, and comprehensive to ensure high-quality analytics (Golfarelli & Rizzi, 2009).

The architecture of a Data Warehouse follows a structured design to facilitate data ingestion, transformation, storage, and analysis. Kimball & Ross, (2013) define four primary components that form the foundation of a Data Warehouse. The first component, source transactions, consists of operational data sources such as Enterprise Resource Planning (ERP) systems, Customer Relationship Management (CRM) platforms, and business applications, which feed raw data into the warehouse. The second component is the ETL (Extract, Transform, Load) system, which extracts data from various sources, performs data cleansing and transformation, and loads it into the warehouse, ensuring data quality and consistency. The third component, the presentation area, organizes data for analytical access, utilizing multidimensional modeling and indexing to optimize query performance. Finally, the Business Intelligence (BI) applications layer provides dashboards, ad-hoc queries, and data mining tools, enabling end-users to extract meaningful insights from stored data (Kimball & Ross, 2013).

A critical aspect of Data Warehouses is dimensional modeling, a widely adopted strategy for structuring data to enhance query performance and usability. Kimball & Ross, (2013) describe two key elements in dimensional modeling: fact tables and dimension tables. Fact tables store measurable business transactions or events, serving as the core dataset for analysis. Dimension

tables, on the other hand, provide multiple perspectives for analyzing fact data, allowing users to segment, filter, and drill into business operations from different angles (Golfarelli & Rizzi, 2009).

By structuring data within these constructs, a Data Warehouse delivers a scalable and structured approach to analytical storage, ensuring fast query performance, intuitive reporting, and improved decision-making processes (Santos & Costa, 2020).

Evolution to Big Data Warehouses

Despite all the benefits introduced by traditional Data Warehouses, such as playing a key role in business intelligence and decision-making in organizations, they were designed with a specific purpose in mind, to handle structured data and batch processing. However, with the emergence of Big Data, these architectures began to show limitations.

As previously analyzed, Big Data has introduced new challenges, requiring Data Warehouses to evolve to support the increasing volume, variety, and velocity of data being generated. Traditional architectures struggled to accommodate semi-structured and unstructured data, as well as the real-time (streaming) analytics needs of modern enterprises. These limitations led to the development of Big Data Warehouses (BDWs), which integrate advanced architectures and distributed processing frameworks to improve scalability, performance, and data integration (Russom, 2016; Santos & Costa, 2020).

A BDW is defined as a high-performance, scalable, and flexible data management system capable of handling large, heterogeneous datasets with low-latency processing (Costa et al., 2019). To overcome the constraints of traditional Data Warehouses, BDWs incorporate several architectural advancements. They leverage flexible and distributed storage, utilizing cloud-based infrastructures, the Hadoop Distributed File System (HDFS), and NoSQL databases to accommodate structured, semi-structured, and unstructured data. Unlike conventional batch-oriented architectures, BDWs support real-time data ingestion and streaming analytics, allowing organizations to process high-velocity data streams efficiently.

Interoperability is another crucial feature of BDWs, as they integrate NoSQL databases, distributed computing frameworks like Apache Spark and Presto, and columnar storage technologies to enhance query performance and analytical processing. These systems are designed to be fault-tolerant and cost-efficient, employing distributed architectures and commodity hardware to ensure high availability while reducing operational expenses (Santos & Costa, 2020).

A Big Data Warehouse is typically divided by two layers: the physical layer and the logical layer, with them being composed of three logical components, each serving a specific function in managing, organizing, and analyzing data (Costa et al., 2019). The first component, Data Collection, Preparation, and Enrichment (CPE), is responsible for ingesting both batch and real-time data, performing cleansing, transformation, and enrichment processes to improve data quality and usability. The second component, Platforms for Data Organization and Distribution, ensures that data is stored in multiple layers, using a combination of file-based and indexed storage systems to optimize performance for various types of workloads. The third layer, Analytics, Visualization, and Access, provides users with advanced querying capabilities through distributed query engines, enabling seamless integration with BI (Business Intelligence) tools and Artificial Intelligence (AI)-driven analytics solutions (Machado et al., 2022b).

Despite their significant advantages, Big Data Warehouses also present challenges that organizations must address. High implementation costs can arise due to the need for specialized data infrastructure, distributed computing resources, and expertise in managing complex data ecosystems. Additionally, data governance complexities become more apparent when integrating diverse data sources, requiring strong metadata management and security frameworks to ensure compliance with regulatory requirements.

Another challenge is latency and performance trade-offs, as handling high-velocity streaming data requires efficient query optimization and workload distribution across multiple computing nodes. Organizations must carefully design data indexing strategies, caching mechanisms, and processing pipelines to minimize latency and enhance analytical performance (Santos & Costa, 2020).

2.3.2 Data Lake

In October of 2010, James Dixon, founder and former Chief Technology Officer (CTO) at Pentaho, introduced the term “Data Lake.” Initially not so accepted, with some viewing it merely as a marketing term associated with Hadoop. Over time, however, the Data Lake concept has gained traction and evolved significantly.

A Data Lake is a centralized repository designed to store vast amounts of data in its raw, unprocessed form, accommodating structured, semi-structured, and unstructured data. This approach allows organizations to ingest data without the need for upfront schema definitions, enabling storage of data "as-is" until it is required for analysis. This flexibility facilitates the

continuous injection of diverse data types into the Data Lake without concern for predefined structures (AWS, 2024).

The primary characteristics of a Data Lake include the storage of data without prior processing and a theoretically infinite storage capacity. For a Data Lake to be effective, it must ensure scalability, high availability, robust data governance, centralized cataloging and indexing, a shared-access model, and support for agile analytics.

Data Lakes and Data Warehouses share the base principle of being both data repositories, but they differ in many other aspects, such as structure and implementation (Khine & Wang, 2018). One other characteristic of Data Lakes, in contrast to Data Warehouses, which employ a "Schema-on-Write" methodology – extracting, transforming, and loading (ETL) data into a predefined schema – Data Lakes utilize a "Schema-on-Read" approach. In this paradigm, data is extracted and loaded in its raw form and transformed only when needed for analysis, following an extract, load, transform (ELT) process (Miloslavskaya & Tolstoy, 2016). This method allows for greater agility, as data is processed and structured at the time of consumption, catering to specific analytical needs.

One characteristic that summarizes the differences between the Data Lake and the Data Warehouse, is relative to the types of data handled by them. While the Data Lake is designed to handle three types of raw data – structured, semi-structured and unstructured – without requiring preprocessing, the Data Warehouse primarily deals with processed and structured data, which has been transformed, cleaned, and optimized for analytical queries (Machado et al., 2022b). Additionally, storage costs in Data Warehouse environments tend to be significantly higher, as they require predefined schemas, optimized storage mechanisms, and extensive ETL processes before data can be used for analysis. This also makes Data Warehouses less agile compared to Data Lakes, which allow for on-the-fly schema application at the moment of data consumption. Another key distinction lies in the targeted user groups of these two systems – while Data Warehouses are typically geared toward business professionals, such as managers and directors, who rely on structured reports and dashboards for decision-making, Data Lakes are more commonly utilized by data scientists and engineers who need direct access to raw data for exploratory analysis, machine learning, and advanced analytics (Khine & Wang, 2018).

Technologically, many Data Lake implementations are built upon frameworks like Apache Hadoop, which allows for the storage of large datasets across distributed clusters. For real-time data processing, frameworks such as Apache Spark and Apache Flink are commonly employed,

enabling the handling of data at varying velocities and the structured storage of heterogeneous data (Khine & Wang, 2018; Databricks, 2023).

A typical Data Lake architecture can be decomposed into multiple layers. The initial layer handles raw transactional data or temporally defined as data within the last six months, followed by the second layer that manages incrementally growing data, or data older than six months that's still used by the organization, and finally, the third layer, that incorporates external information and archived data, not as frequently used. This stratification ensures efficient data management and retrieval based on usage patterns and data relevance (Miloslavskaya & Tolstoy, 2016).

All these flexibility features can be a valuable asset for organizations, but they can also become a significant challenge if not properly managed. As defined by Khine & Wang, (2018), when issues such as the lack of governance in a Data Lake arise, they can lead to what is commonly referred to as a Data Swamp – a disorganized and unmanaged repository that diminishes the value of stored data. To address this challenge, multiple research initiatives and organizations have focused on enhancing and optimizing Data Lake services. Various solutions have been proposed, including mechanisms for extracting and summarizing structural metadata from data sources to improve metadata management and prevent governance failures. Additionally, open-source services have been developed, such as Representational State Transfer (REST) Application Programming Interfaces (APIs), which enable efficient indexing and organization of metadata within a Data Lake. These advancements help mitigate the difficulties caused by heterogeneous data sources, ensuring better data discoverability, consistency, and usability across the Data Lake ecosystem.

To summarize, as Data Lake serves as a vast repository capable of accepting structured, semi-structured and unstructured data, offering scalability and flexibility for diverse analytical needs, scaling theoretically to infinite amounts of raw information. Unlike traditional Data Warehouses, which rely on predefined schemas and structured processing, Data Lakes provide a more adaptable and cost-efficient approach to handling large and complex datasets. This capability makes them particularly suitable for high-volume, high-velocity data environments. However, without proper governance mechanisms, a Data Lake can quickly deteriorate into a Data Swamp, where poor metadata management, lack of structure, and uncontrolled data growth hinder usability and strategic value.

2.3.3 Data Lakehouse

The growing need for flexible, scalable, and high-performance data analytics platforms has led to the emergence of the **Data Lakehouse** paradigm, an architectural model that unifies the low-cost, schema-flexible storage of Data Lakes with the management, reliability, and performance features traditionally associated with Data Warehouses. Initially promoted by industry platforms such as Databricks and later formalized within academic literature, the Lakehouse attempts to reconcile the long-standing divide between lakes and warehouses by supporting both transactional and analytical workloads in a single, integrated system (Armbrust et al., 2021).

According to Armbrust et al. (2021), a Data Lakehouse is characterized by its ability to provide Atomicity, Consistency, Isolation, and Durability (ACID) transactions, schema enforcement and evolution, granular governance controls, and performance optimizations on top of scalable cloud object storage. Unlike conventional Data Warehouses, which impose rigid schemas and require extensive ETL processes, or traditional Data Lakes, which often lack strong governance and data quality guarantees, the Lakehouse model adopts a metadata-driven approach that allows raw, semi-structured, and structured data to coexist seamlessly, commonly using formats such as Apache Parquet or Delta Lake (Armbrust et al., 2020).

Architecturally, a Data Lakehouse relies on cloud object storage as the underlying persistence layer, complemented by a transaction and metadata management system that enables features such as time travel, schema evolution, and ACID guarantees. Technologies like Delta Lake, Apache Hudi, and Apache Iceberg exemplify mature implementations of this layer. On top of this foundation, the Lakehouse offers unified data access, enabling SQL analytics, machine learning, and streaming operations over the same data without requiring duplicated pipelines. This integration eliminates the traditional division between OLAP platforms and data science environments, enabling a consolidated and efficient analytical ecosystem.

The Lakehouse model therefore simplifies data architectures by reducing system fragmentation and the need for complex synchronization procedures across multiple platforms. It supports a diverse set of analytical and operational use cases, from real-time processing to advanced AI and machine learning workflows, and does so while preserving flexibility through both schema-on-read and schema-on-write capabilities. These characteristics make it especially relevant for organizations operating in dynamic environments where data variety and volume evolve rapidly.

Despite its advantages, the Data Lakehouse introduces several practical challenges. Managing metadata and ensuring optimal query performance at scale may lead to performance overheads. Deploying open-source Lakehouse engines within hybrid or multi-cloud environments can increase configuration and operational complexity. Furthermore, although the model has gained substantial momentum in industry, it remains relatively young in terms of academic standardization and empirical validation. Nevertheless, recent research suggests that Lakehouse architecture provides superior cost-efficiency and query performance compared with traditional warehouse-centric models when operating at petabyte scale in cloud-native infrastructures.

In summary, the Data Lakehouse represents a significant evolution in modern data architecture by combining the governance, quality, and performance capabilities of a Data Warehouse with the flexibility and cost-effectiveness of a Data Lake. It provides a unified and scalable foundation for data management and analytics, positioning itself as a central component of contemporary data ecosystems and an enabler of advanced data-driven innovation.

2.3.4 Comparison between paradigms

The following section presents a comparative table that summarizes a set of defining characteristics, namely Governance, Scalability and Operational Efficiency, Data Integration and Interoperability, Security, Analytical Performance, Processing Approach, Supported Data Types, Target Users, and Cost Considerations, across the Data Warehouse, Data Lake, and Data Lakehouse paradigms.

Table 1: Comparison Between Data Management Paradigms

Feature	Data Warehouse	Data Lake	Data Lakehouse
Governance	Mature and centralized governance with structured data curation and metadata control (Kimball & Ross, 2013).	Limited governance unless extended; risk of becoming a “Data Swamp” if not properly managed (Khine & Wang, 2018).	Incorporates governance mechanisms such as schema enforcement and fine-grained access via metadata layers (Armbrust et al., 2021).
Scalability and Operational Efficiency	Limited scalability; optimized for structured, batch-oriented analytics; high operational overhead.	High scalability due to distributed storage and schema-on-read, but lacks inherent compute optimization.	Highly scalable; combines low-cost storage with performance optimizations such as caching and indexing.

Data integration and Interoperability	Effective for structured data; requires extensive ETL and predefined schemas (Golfarelli & Rizzi, 2009).	Highly flexible; supports structured, semi-structured, and unstructured data using ELT pipelines.	Unified approach for batch, real-time, and diverse data types; supports both schema-on-read and schema-on-write.
Security	Robust and mature security frameworks, including role-based access and auditing mechanisms.	Varies by implementation; often lacks centralized control unless complemented with additional tools.	Leverages modern security features from both lake and warehouse paradigms; integrates encryption and policy layers.
Analytical Performance	High for OLAP and structured reporting; low latency for predefined queries.	Poor for complex queries unless tuned; relies on external compute engines (Spark, Flink).	High performance with support for real-time analytics and machine learning (ML) workloads via integrated engines (Delta, Iceberg).
Processing Approach	Schema-on-write; rigid transformation pipelines.	Schema-on-read; transformation applied at query time.	Hybrid; supports both schema-on-write for governance and schema-on-read for flexibility.
Data Types Supported	Structured data only.	Structured, semi-structured, and unstructured.	All types supported; integrated with metadata to ensure usability and consistency.
Target Users	Business analysts and decision-makers using BI dashboards.	Data engineers and scientists performing exploration, ML, or streaming analytics.	Both technical and business users; unifies data science and BI access.
Cost Considerations	Higher costs due to optimized storage engines and rigid infrastructure requirements.	Storage-efficient; lower upfront cost but high data wrangling effort.	Balanced cost-performance trade-off with cloud-native elasticity.

Overall, this comparison illustrates the evolution from rigid, highly governed architectures towards more flexible and scalable platforms, and ultimately towards hybrid approaches that seek to balance both. The Data Lakehouse paradigm represents this convergence, addressing

the core limitations of Data Warehouses and Data Lakes while supporting modern analytical and governance demands.

2.4 Decentralized Data Management Paradigms

In contrast, decentralized data management distributes data ownership, storage, and governance responsibilities across multiple domains or business units. This model is particularly well-suited for organizations with diverse operational needs, where business agility, scalability, and domain-specific autonomy are prioritized. Instead of relying on a single governing body, decision-making power is distributed among different entities, allowing departments to manage their own data independently while ensuring interoperability with other systems (Rülicke et al., 2024).

Governance:

Unlike centralized models, decentralized systems distribute decision-making power across multiple departments or business units, allowing each domain to define its own data policies and access control mechanisms. This ensures greater flexibility and responsiveness to domain-specific requirements but also necessitates a federated governance framework to prevent inconsistencies and inefficiencies (Gelhaar & Otto, 2020). Access rights are managed at a domain level, with metadata registries and data contracts used to facilitate controlled data sharing between different units of the business. While compliance practices are often standardized at a high level, their implementation may vary across business domains, introducing challenges in ensuring regulatory alignment.

Scalability and Operational Efficiency

One of the key advantages of decentralization is its ability to scale dynamically. Unlike centralized models, where data must be funneled through a single control point, decentralized systems allow each domain to manage its own data pipelines, ensuring efficient storage, processing, and retrieval (Islam & Apu, 2024). However, this distributed architecture requires robust data-sharing frameworks to maintain efficiency across multiple domains.

Data Integration and Interoperability

While decentralization enhances operational autonomy, it introduces challenges in data integration and interoperability. Without a common governance framework, data silos may

emerge, leading to fragmented and inconsistent data landscapes. Organizations must implement metadata management, semantic interoperability, and standardized APIs to facilitate cross-domain data exchange (Ribeiro et al., 2024).

Security

Decentralized approaches require stronger security mechanisms to prevent unauthorized access across multiple independently managed domains. Without centralized security controls, ensuring regulatory compliance across all units becomes more complex (Rülicke et al., 2024). Techniques such as architectures, decentralized identity management, and federated security models are often necessary to maintain data integrity and compliance.

Despite this, decentralized models are becoming increasingly popular in industries requiring agility and scalability, such as technology, e-commerce, and multinational corporations with diverse business units. By allowing business domains to retain control over their own data, organizations can improve decision-making speed, innovation, and adaptability to market changes.

2.4.1 Data Mesh

The initial contributions to Data Mesh came from Dehghani, (2019), who presented it as a disruptive and innovative paradigm that structures data into nodes with the primary objective of decentralizing data ownership and architecture. As discussed in previous sections, traditional monolithic architectures, such as Data Warehouses and Data Lakes, often lead to performance bottlenecks due to centralized governance and data processing. While Data Lakes have introduced significant improvements in scalability and flexibility, they still struggle with inefficiencies related to centralized control and data access management.

To address these limitations, Data Mesh advocates for a distributed, domain-driven model, where individual teams own and manage their data as a product, ensuring interoperability and governance through standardized infrastructure principles (Dehghani, 2019; Barr, 2020). By shifting from centralized data management to a federated approach, Data Mesh aims to enhance scalability, flexibility, and efficiency, by reducing bottlenecks in data workflows.

In a 2020 article retrieved from (ThoughtWorks, 2020), Data Mesh's three core principles and two critical challenges were outlined. The first core principle, one already mentioned above, is domain-oriented data ownership, with this meaning that the responsibility for data management

is assigned to each individual business domain, guaranteeing that those with the most knowledge about the data are accountable for its governance and quality. The second principle introduces the idea of data as a product, advocating that datasets should be treated with the same level of rigor as any other product in an organization – prioritizing quality, usability, and discoverability. The third principle highlights the self-service nature of the Data Mesh infrastructure, which enables domain teams to manage and process their data without relying on centralized data engineering teams.

In addition to these core principles, ThoughtWorks (2020) also highlights two major challenges in the implementation of Data Mesh: technological adaptation and organizational restructuring. This involves the shift from the older monolithic platforms to decentralized architectures and domain-specific data products, while from an organizational standpoint, organizations must shift their policies about data ownership, workflows, and team responsibilities, to a distributed, cross-functional model.

An alternative perspective is provided by (Max & Arif, 2020), who offer a practical and streamlined perspective on Data Mesh, emphasizing data ownership over its technological aspects. Drawing from their real-world experience, they identify three fundamental pillars that underpin a successful Data Mesh: product thinking, domain-driven distributed architecture, and infrastructure as a platform. They argue that effective implementation hinges on treating data as a first-class product, structuring architectures around business domains, and empowering teams with a self-sufficient data infrastructure.

2.4.2 Data Spaces

Now that all foundational concepts, that serve as a base to understand the evolution of Data Management Architectures and the need for their transformation, have been established, we now transition to the core focus of this dissertation. The following section will dive into the concept of Data Spaces, exploring their fundamental principles, the ecosystems in which they exist, and their broader implications in modern data management.

The concept of Data Spaces was first introduced in the computer science literature in 2005 as an innovative approach to data integration. In their seminal paper, Franklin et al., (2005) proposed "dataspaces" as a new abstraction for information management, aiming to provide base functionality over all data sources, regardless of how integrated they are.

Building upon this foundation, Halevy et al., (2006) further elaborated on the principles of dataspaces, emphasizing the need for Data Space Support Platforms (DSSPs) that offer services without requiring full semantic integration of the sources.

The concept kept evolving and in 2017, the International Data Spaces Association (IDSA) was established. The IDSA is dedicated to developing standards and frameworks to facilitate secure and sovereign data sharing among organizations. Their work includes the development of the IDS Reference Architecture Model, already cited in this dissertation, which provides guidelines for creating data spaces that ensure data control and compliance.

2.4.3 Comparison between paradigms

To consolidate the distinctions discussed in the previous sections, Table 2 presents a comparative analysis between the Data Mesh and Data Spaces paradigms across a set of defining dimensions, including governance, interoperability, scalability, security, and target users. The comparison aims to clarify how, despite certain conceptual overlaps, particularly regarding decentralization and federated governance, where the two paradigms address fundamentally different problem spaces.

Table 2: Comparison Between Data Mesh and Data Spaces Paradigms

Feature	Data Mesh	Data Spaces
Governance	Federated governance model where each domain governs its own data product; enforced via platform-wide policies and standards (Dehghani, 2019).	Federated and policy-based governance driven by common standards (e.g., IDSA); enables data sovereignty and compliance (IDSA, 2019).
Scalability and Operational Efficiency	High scalability through distributed architecture; operational responsibility is shifted to domain teams (Barr, 2020).	Scalable across organizations or ecosystems; encourages interoperability while maintaining autonomy over data (Franklin et al., 2005).
Data integration and Interoperability	Standardized APIs and schemas ensure interoperability between decentralized data products; relies on shared contracts (Max & Arif, 2020).	Interoperability is central; relies on common ontologies, connectors, and metadata vocabularies (IDSA 2019; Halevy et al., 2006).
Security	Varies by domain; requires enforcement of centralized security principles via platform	Strong emphasis on secure and sovereign data exchange; supports identity management, auditability,

	tooling; depends on maturity of self-service infra.	and secure connectors (IDSA, 2019).
Analytical Performance	Performance varies by implementation; enables local optimization per domain; avoids monolithic bottlenecks.	Depends on the underlying platforms; promotes modular and extensible components for analytic services across domains.
Processing Approach	Domain teams define ingestion and transformation pipelines; both batch and streaming are supported with autonomy.	Focuses on federated data sharing rather than centralized processing; relies on connectors for selective data exchange.
Data Types Supported	All data types; heterogeneity handled within each domain via fit-for-purpose tools.	Designed to enable sharing of diverse data assets across verticals, including structured and unstructured sources.
Target Users	Cross-functional teams (data producers and consumers) within an organization; data product owners, engineers, analysts.	Independent organizations and institutions participating in shared ecosystems (e.g., public-private collaborations).
Cost Considerations	Reduced central infrastructure costs; requires investment in internal domain capabilities and tooling for autonomy.	Can be complex to implement and maintain across institutions; cost-effective when infrastructure and governance are shared.

While Data Mesh is primarily conceived as an intra-organizational approach focused on scaling analytics and data ownership within a single enterprise, Data Spaces are designed to support inter-organizational data sharing across autonomous and independently governed participants. This distinction is reflected not only in governance and interoperability mechanisms, but also in the role of standards, trust frameworks, and policy enforcement. The comparison therefore highlights that Data Spaces should not be interpreted as a direct extension of Data Mesh, but rather as a complementary paradigm addressing challenges that arise beyond organizational boundaries.

2.5 In Depth Look of Data Spaces

Having positioned Data Spaces in relation to other contemporary data management paradigms, this section provides a more detailed examination of the concept. The objective is to clarify the foundational characteristics of Data Spaces by introducing their definition, core principles, and structural components, as discussed in both academic literature and European policy initiatives.

This in-depth analysis establishes the conceptual basis required for the architectural and implementation-oriented discussions developed in the subsequent chapters.

2.5.1 Definition of Data Spaces

Starting by the definition of Data Spaces accordingly to the European Commission in Common European Data Spaces (European Commission, 2023), they are a decentralized framework designed to facilitate secure and trustworthy data sharing and exchange among various participants within a data ecosystem. Unlike traditional centralized data architectures, Data Spaces emphasize data sovereignty, allowing data providers to retain control over their data and determine access conditions. This approach ensures that data sharing complies with relevant laws and regulations, promoting fair treatment for all involved parties (European Commission, 2023).

In essence, a Data Space provides a structured environment where data from diverse sources and formats can be integrated, linked, and made accessible to authorized data consumers. This integration is achieved through commonly agreed-upon principles, standards, and practices that ensure interoperability and trust among participants. The decentralized nature of Data Spaces means that data remains at its source and is shared only when necessary, preserving the sovereignty of data providers (Nagel et al., 2022).

The concept of Data Spaces has been further developed by organizations such as the International Data Spaces Association (IDSA), which defines a Data Space as a secure and standardized digital infrastructure enabling trusted data exchange and data-based services among various stakeholders (Otto et al., 2019). The IDSA's framework supports data sovereignty, ensuring that data owners maintain control over their data and can specify usage conditions.

2.5.2 Data Spaces Environment

Traditional data architectures, such as Data Warehouses, Data Lakes, and even Data Mesh, primarily focus on data management within a single organization. These architectures are designed to centralize, organize, and process data in a way that benefits internal operations, enabling better business intelligence, analytics, and decision-making. While Data Mesh introduces decentralization within an enterprise by distributing data ownership across different domains, they were mainly thought to operate within the boundaries of a single organization. In contrast, Data Spaces are designed to extend beyond the organizational level, facilitating

secure, standardized, and sovereign data exchange between multiple organizations. They enable inter-organizational collaboration, allowing different entities to share and utilize data while maintaining control, privacy, and compliance (Otto et al., 2019). Unlike previous architectures, which optimize data management internally, Data Spaces are designed to bridge data silos across enterprises and industries, ensuring trust, interoperability, and sovereignty in data-sharing ecosystems. These spaces are particularly relevant in sectors such as healthcare, finance, energy, and manufacturing, where cross-organizational collaboration is crucial for innovation, regulatory compliance, and operational efficiency.

Unlike traditional architectures, which are built as monolithic or unified systems, Data Spaces function as federated ecosystems composed of multiple autonomous Data Products. Each Data Product within a Data Space may adopt a different architectural model, such as a Data Warehouse for structured analytics, a Data Lake for storing heterogeneous data, or a Data Lakehouse for combining agility with governance (Otto et al., 2019). This modularity allows organizations to maintain internal architectural autonomy, while participating in external standardized exchanges via shared APIs, metadata vocabularies, and interoperability protocols (IDSA, 2019).

In this context, Data Spaces serve as macro-level integration environments, where the emphasis lies not on the internal structure of each data node, but on the cross-domain data sharing mechanisms that ensure trust, sovereignty, and compliance. As Halevy et al. (2006) note, such environments do not require full semantic integration upfront but instead promote gradual and context-aware alignment based on common governance principles and interoperability standards.

Driven by initiatives such as the European Union's Data Strategy, Data Spaces are increasingly seen as the foundation for future digital ecosystems. They enable organizations to collaborate effectively while upholding privacy, competition laws, and sector-specific regulations through a federated yet cohesive governance structure. This approach empowers data providers to define how, when, and under what conditions their data is accessed and used, ensuring that collaboration does not compromise sovereignty, security, or legal compliance.

2.5.3 Core Principles of Data Spaces

According to the International Data Spaces Association (IDSA, 2019), Data Spaces operate based on **five core principles** that ensure secure, interoperable, and sovereign data exchange.

- **Data Sovereignty**: The principle of data sovereignty ensures that participants retain full control over their data, allowing them to determine how, when, and with whom their data is shared.
- **Interoperability**: It plays a crucial role in Data Spaces by establishing standardized protocols that enable compatibility across different data platforms and technologies, facilitating seamless integration within diverse ecosystems.
- **Decentralized Approach**: Unlike centralized databases, Data Spaces follow a decentralized approach, where data remains physically stored with its owner until it is exchanged with a trusted participant, preserving ownership and security.
- To ensure **Transparency and Security**, trust and governance mechanisms are enforced to ensure compliance, data protection, and secure transactions among all participants.
- **Federated Governance**: Lastly, this principle allows decision-making rights and policies to be distributed among participants, rather than imposed by a central authority, promoting collaborative and fair data management (Otto et al., 2019; IDSA, 2019).

This decentralized yet structured approach enables organizations to participate in data ecosystems while adhering to regulatory and industry-specific standards, fostering trust and transparency in inter-organizational data collaboration.

2.5.4 The Need for Data Spaces: Why They Emerged

The main driving factors for the emergence of Data Spaces, according to IDSA, were the challenges faced by organizations regarding data sharing, governance and interoperability. As explained above, Data Warehouses and Data Lakes were primarily designed for internal enterprise data management and struggle to facilitate secure, federated data exchange between different entities. In addition, organizations often store their data in isolated silos, leading to fragmented data ecosystems where interoperability and seamless data integration become increasingly complex (Otto et al., 2019).

Despite the increasing relevance of Data Spaces, the field is still maturing, with limited practical guidance and architectural approaches available for organizations aiming to implement these ecosystems (Curry & Tuikka, 2022).

Regulatory frameworks, such as the General Data Protection Regulation (GDPR) and the Data Governance Act (DGA), have further reinforced the need for greater control over data usage policies, ensuring that organizations comply with legal, ethical, and security standards when

sharing data (Otto et al., 2019). Another driving factor behind Data Spaces is the rising demand for industry-specific data exchanges. Sectors such as healthcare, energy, and automotive require tailored data-sharing frameworks that enable secure and domain-specific collaboration, ensuring that data can be exchanged while preserving regulatory compliance, sovereignty, and trust among stakeholders (Rülicke et al., 2024).

2.5.5 Key Components of a Data Space

The Reference Architecture Model (IDS-RAM), developed by the International Data Spaces Association (IDSA) and Otto et al., (2019) structures Data Spaces into five core layers, ensuring a standardized and modular approach to data exchange. These layers define the roles, functions, processes, information models, and system requirements that facilitate secure and interoperable data sharing among participants.

At the Business Layer, the roles, responsibilities, and interactions within a Data Space are established, encompassing data providers, consumers, and intermediaries. This layer defines how participants engage within the ecosystem, specifying their main activities and the relationships between them. Moving to the Functional Layer, this component outlines the functional requirements of a Data Space in a technology-independent manner. It specifies the capabilities and services required for data sharing, governance, and interoperability, ensuring that all stakeholders can effectively and securely exchange data.

The Process Layer provides a dynamic view of the architecture, describing the lifecycle of data transactions, including data onboarding, exchange, and access control. This ensures that all interactions between components are well-defined, transparent, and manageable. The Information Layer serves as the foundation for semantic interoperability, providing a common vocabulary and metadata framework that allows participants to establish standardized agreements and contracts. This layer ensures that data is structured in a way that facilitates consistent interpretation and collaboration across different domains.

At the System Layer, the architecture defines the logical decomposition of software components, focusing on aspects such as integration, configuration, deployment, and extensibility. This layer ensures that all technical security features are in place, enabling safe and compliant data transfers between participants. Additionally, the IDS Connector serves as a core technical component, facilitating secure data transactions, identity management, and policy enforcement. It acts as the primary interface through which participants engage with the

Data Space, ensuring that all exchanges are conducted securely and in accordance with established governance policies.

2.5.6 Data Governance in Data Spaces

Governance plays a crucial role in Data Spaces, ensuring trust, compliance, and efficient data management across decentralized ecosystems. Unlike traditional centralized governance models, Data Spaces adopt a federated governance approach, allowing decision-making to be distributed among participants while maintaining global interoperability and regulatory compliance (Otto et al., 2019).

Within a Data Space, governance is structured around four key roles that help maintain data integrity, security, and controlled access. The Data Owner or Provider holds the responsibility for defining data usage policies, access rights, and quality standards, ensuring that data sovereignty is upheld and that data providers retain full control over access and usage conditions. The Data Consumer is responsible for utilizing shared data while adhering to governance policies and security requirements, ensuring that all data interactions align with legal and ethical standards. To facilitate efficient data transactions and metadata management, the Broker Service Provider acts as an intermediary, ensuring seamless and secure data exchange between different stakeholders. Overseeing the entire governance framework, the Governance Body establishes standards, compliance mechanisms, and trust protocols to regulate the ecosystem, ensuring that all participants adhere to a set of global policies that promote transparency and secure collaboration (Otto et al., 2019).

By leveraging a federated governance model, Data Spaces ensure that decision-making remains decentralized, granting participants autonomy over their data while still complying with international data-sharing standards. This approach fosters trust, transparency, and secure collaboration, enabling organizations to exchange and utilize data efficiently while maintaining full control over their information assets.

3. PROPOSAL OF AN APPROACH FOR INTER-ORGANIZATIONAL DATA ARCHITECTURE: DATA SPACES

As established in the previous chapter, the evolution of data management paradigms has led to the emergence of Data Spaces as a federated, secure, and interoperable framework for inter-organizational data exchange. While traditional architectures have focused on centralizing or internally distributing data assets, Data Spaces introduce a macro-level approach to data integration, prioritizing data sovereignty, trust, and governance across heterogeneous participants and infrastructures.

This third chapter proposes an approach for designing and implementing an inter-organizational Data Space, structured into three models: a domain model, a conceptual architecture, and a technological architecture. Together, these models specify the conceptual and technological components that constitute a Data Space, thereby establishing actionable knowledge to support its design and implementation across contexts. First, the Data Space domain model is introduced and detailed, including all constructs that conceptually should exist in a trustworthy and sovereign data-sharing environment. Next, the conceptual architecture identifies generic functional components and their interactions, aligning with European policy, the IDS Reference Architecture, and Data Spaces maturity perspectives. Finally, to assist the translation of the conceptual architecture into a feasible implementation, the technology architecture is presented, mapping conceptual components to families of technologies and platform capabilities.

3.1 Data Spaces Domain Model

The elaboration of a domain model is a fundamental step in problem investigation, as it represents the conceptual classes and their associations that characterize the real world rather than software components (Larman, 2004). By abstracting the key entities, relationships, and constraints within a given domain, such a model supports a shared understanding of the system's conceptual structure and provides a stable foundation for the subsequent phases of architectural design and technological implementation. It thus serves as an analytical bridge between theoretical knowledge and its practical realization within the proposed data architecture.

Accordingly, the specification of an approach for the design and implementation of Data Spaces starts with the construction of its domain model. The model that is presented in this dissertation is the result of the extensive literature review that was conducted on Data Spaces and related paradigms, the analysis of the policy and regulatory guidance for European data sharing initiatives, the Data Spaces common policy framework and the critical reading of empirical and conceptual studies that highlight issues related to trust, governance, and organizational maturity in data ecosystems.

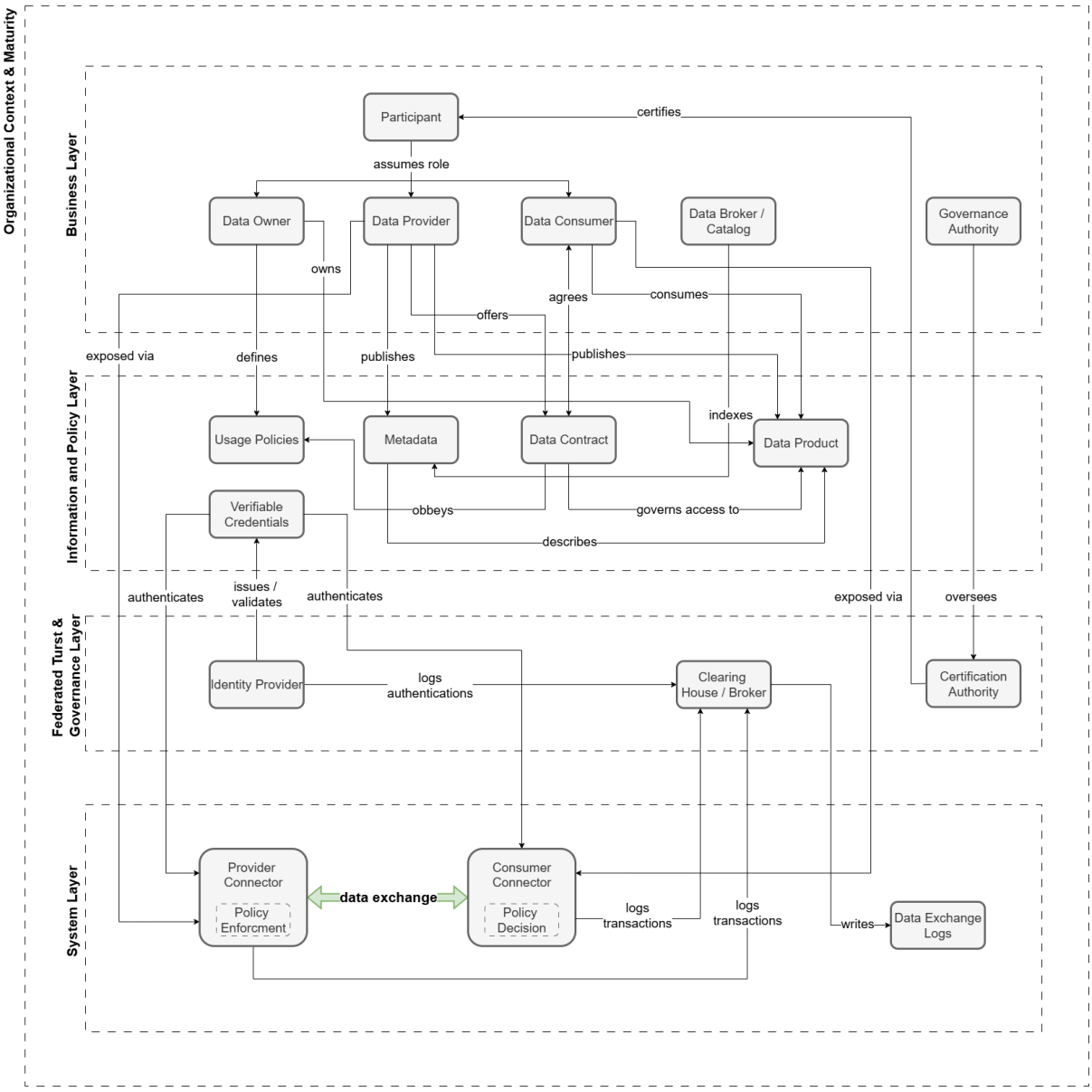


Figure 2: Data Spaces Domain Model

3.1.1 Overview and Guiding Principles

The Data Spaces Domain Model Diagram presented above, in Figure 2, was developed by the author as part of the proposed contribution. It is not an adaptation of a specific external diagram,

although its structure and conceptual organization are based on the principles of the IDS Reference Architecture Model and by the broader Data Spaces literature. Through its analysis, several defining characteristics can be identified: Data Sovereignty, Interoperability, Decentralization, Transparency and Security, and Federated Governance (European Commission, 2023). Unlike intra-organizational models, such as the Data Mesh, which primarily promotes decentralization within a single enterprise, the Data Space expands the scope of decentralization to the inter-organizational and cross-sector level, establishing a federated ecosystem where independent participants collaborate under shared governance and trust frameworks (Otto et al., 2019).

Within this environment, **Data Sovereignty** ensures that each participant maintains full control over its data assets, determining access conditions and permissible uses. **Interoperability** enables seamless integration across heterogeneous systems, through shared technical and semantic standards. **Decentralization** guarantees that data remains at its original source, eliminating dependency on centralized repositories and reducing risks associated with single points of failure. **Transparency and Security** are upheld through certified components, verifiable credentials, and auditable data exchanges that ensure compliance and accountability. Lastly, **Federated Governance** distributes decision making and policy enforcement across participants, rather than a central authority, fostering fairness, trust and collaboration across the entire data sharing ecosystem. (IDSA, 2024).

The model is organized using layer-oriented abstractions inspired by the IDS Reference Architecture Model (IDSA, 2019), structuring actors, policies, trust services, technical enforcement mechanisms, and organizational readiness.

3.1.2 Core Actors and Roles

The foundation of the Data Spaces lies in the interaction of several participants, each assuming a specific role that defines its responsibilities and relations within the ecosystem. According to the IDS-RAM (IDSA, 2019), these **actors** form the **Business Layer**, where the primary economic, legal, and operational functions of the Data Space are established. The **Data Owner** represents the legal entity holding the rights over a data asset and is responsible for defining the corresponding usage policies, determining the conditions under which data can be accessed, shared or reused. Building upon the rights, the **Data Provider** operationalizes data publication by exposing datasets, metadata, and access endpoints, ensuring that information can be discovered and securely accessed through standardized mechanisms. Complementarily, the

Data Consumer requests and utilizes these data assets under formally established **contracts**, adhering to the defined policies and ensuring lawful and ethical use.

The **Data Broker** or **Catalog** plays a crucial role in intermediating dataset discovery, matching supply and demand across multiple participants, and maintaining metadata registries that describe available data assets and their associated conditions. Finally, the **Governance Authority** functions as a supervisory and coordinating entity, ensuring that participants comply with shared rules and certification requirements, by promoting interoperability and policy harmonization across the entire ecosystem.

These actors operate under a Federated Governance Model, in which authority and decision making are distributed among multiple participants, rather than concentrated in a single organization. This decentralized arrangement allows for higher autonomy and accountability while maintaining coherence through shared governance principles and interoperability standards.

In practical terms, these roles manifest differently across domains. If we look at an example from the energy domain, data providers such as grid operators and utilities share operational data with data consumers like energy traders and research entities through brokers or market platforms, under the oversight of regulatory authorities acting as governance bodies. (Rülicke et al., 2024) Similarly, in the healthcare sector, hospitals and healthcare providers may act as data providers, research institutions and public agencies as consumers, and health data hubs serve as intermediaries ensuring compliance and interoperability under the supervision of national health or data protection authorities.

3.1.3 Information and Policy Layer

Positioned above the business layer, the **Information and Policy layer** defines the conceptual and regulatory foundations that ensure trust, compliance, and interoperability, translating legal and organizational principles into actionable mechanisms that regulate data interactions. (Halevy et al., 2006) Within this dimension, metadata plays a central role, by providing the contextual information that makes datasets discoverable, understandable, and interoperable across domains. By describing data assets through standardized schemas, providing data lineage, and semantic annotations, metadata supports both technical alignment and governance transparency. Complementing this, **usage policies** establish the terms and conditions that govern how data can be accessed and utilized by consumers defining rights, obligations and restrictions in alignment with the Data Governance Act. (Regulation (EU) 2022/868, 2022)

These policies are operationalized through machine-readable rules that can be automatically enforced by technical components, ensuring that all exchanges adhere to legal and ethical requirements.

Building upon these elements, **data contracts** formalize the bilateral agreements between providers and consumers, defining the scope of data usage, licensing conditions, and compliance obligations. These contracts function as the legal instruments that bind participants to specific terms, while maintaining traceability and accountability throughout the data lifecycle. To guarantee semantic coherence, the **semantic interoperability layer** provides shared vocabularies and ontologies that enable consistent interpretation of data across heterogeneous systems and sectors. This semantic alignment allows for the integration of diverse data sources without compromising their contextual meaning, a crucial requirement for multi-domain collaboration. Finally, **verifiable credentials** strengthen the trust framework by providing cryptographic attestations of identity, certification, and authorization. Through these credentials, issued by trusted authorities within the governance framework, participants and components can verify the legitimacy of their counterparts before engaging in data transactions. Collectively, these constructs establish a normative and semantic infrastructure that ensures every data exchange within the Data Space is self-describing, compliant, and auditable. As emphasized in reflections on the Data Governance Act and the role of data intermediaries, this layer operationalizes the principles of the Data Governance Act (Regulation (EU) 2022/868, 2022) by embedding transparency, neutrality, and accountability into the technical design of intermediation services. It thereby transforms abstract regulatory requirements into enforceable data-sharing mechanisms, bridging the gap between legal governance and technical interoperability, and reinforcing the overarching values of trust and data sovereignty that underpin the Data Space ecosystem.

3.1.4 Federated Trust and Governance Framework

At the core of every Data Space lies the **Federated Trust and Governance Framework**, the structural element that distinguishes this paradigm from conventional data integration architectures. This backbone operationalizes the notion of a federated trust framework, which enables autonomous organizations to exchange data securely and transparently while maintaining full sovereignty over their assets. It is composed of interconnected governance and assurance mechanisms that collectively establish identity, accountability, and compliance across all participants. Within this framework, the **Identity Provider** plays a pivotal role by

authenticating participants and validating the credentials of both users and system components, ensuring that only certified entities can engage in data transactions. This process is complemented by the **Clearing House** or **Broker**, which guarantees transparency and auditability through immutable logs that record every data exchange, supporting accountability, traceability, and dispute resolution. Overseeing these mechanisms, the **Governance Authority** defines overarching policies, enforces certification schemes, and ensures alignment with legal and regulatory frameworks such as the Data Governance Act (Regulation (EU) 2022/868, 2022) and the IDS certification framework (IDSA, 2019).

This governance structure enforces the principles of transparency, neutrality, and accountability, serving as the institutional foundation of the ecosystem. As (Otto, 2011) highlights, effective data governance requires not only internal control mechanisms but also the extension of governance structures beyond organizational boundaries. The federated governance of Data Spaces thus represents the evolution of traditional corporate data governance into an inter-organizational model, where trust is distributed and institutionalized through standardized protocols, certification bodies, and regulatory oversight. Through this backbone, each data exchange within the Data Space becomes secure, verifiable, and legally compliant, ensuring that all participants operate within a shared framework of trust and mutual assurance.

3.1.5 System Layer: Connectors and Sovereign Exchange

At the technological level, the **System Layer** represents the operational core through which the conceptual and governance principles of the Data Space are technically enacted. It is within this layer that sovereign data exchange takes place, implemented through standardized and certified components that translate contractual and policy-level agreements into executable technical controls. The central components of this layer are the **Provider and Consumer Connectors**, which function as secure gateways enabling the exchange of data between participants. Each connector enforces the policies and data contracts defined in the higher layers. By maintaining standardized interfaces and protocols, these connectors guarantee interoperability across heterogeneous systems, while preserving the autonomy of each participant's internal infrastructure.

Embedded within the connectors are **usage control mechanisms**, typically represented by Policy Decision and Enforcement Points (PDP/PEP), which dynamically regulate access and purpose limitations during the data lifecycle. These mechanisms continuously verify whether

data usage adheres to the defined contractual obligations, applying restrictions in real time to prevent unauthorized processing or misuse. Such functionality is critical to maintaining **data sovereignty**, as it allows providers to retain ongoing control over their data even after it has been shared.

As outlined in the IDS Reference Architecture Model (IDSA, 2019; Otto et al., 2019), the system layer provides the technical realization of the Information, Process, and Business layers, integrating them into a unified operational environment. The connectors act as the tangible manifestation of the Data Space's federated architecture, enabling decentralized data exchange while maintaining global governance coherence. In this sense, the system layer exemplifies what (L. Nagel & Lycklama, 2022) describe as a **"decentralized yet governed infrastructure"**, where interoperability and compliance are achieved through standardized components rather than central control. Ultimately, this layer transforms the abstract governance principles of the Data Space: **sovereignty, transparency, and interoperability**, into enforceable, technically secure mechanisms that underpin trusted and legally compliant data exchange.

3.1.6 Organizational Context and Maturity

Encircling the technical and conceptual core, the **Organizational Context and Maturity Layer** constitutes the socio-organizational foundation that enables the sustainable and effective operation of a Data Space. This layer, as outlined in the Data Spaces Maturity Model (Curry & Tuikka, 2022), captures the key capabilities that organizations must develop to participate successfully in federated data-sharing ecosystems. At its essence, this context ensures that technological interoperability is complemented by organizational alignment, strategic intent, and governance competence, transforming technical infrastructures into enduring, value-driven collaborations.

Within this framework**,** organizational strategy plays a central role, as the objectives of data sharing must be consistent with the organization's broader business goals and regulatory responsibilities. Value creation mechanisms define how shared data generates economic and societal benefits, such as innovation, improved efficiency, or enhanced public services, ensuring that participation in the Data Space is mutually advantageous. Effective data lifecycle management and stewardship guarantee that datasets remain of high quality, traceable, and ethically managed throughout their use, fostering trust among participants. Robust technological infrastructure underpins this by ensuring the availability of secure, scalable, and

interoperable environments that support continuous data exchange. Equally vital are the people and cultural dimensions, which encompass the skills, collaboration practices, and trust relationships necessary for cross-organizational engagement. Finally, governance and assurance mechanisms maintain ongoing compliance, certification, and accountability, aligning local operational practices with shared policies and European data governance frameworks.
By integrating these dimensions, the outer layer underscores that **Data Spaces are socio-technical systems** whose sustainability relies as much on human and institutional maturity as on technical design. For organizations seeking to join or establish a Data Space, this means developing not only interoperable infrastructures but also strategic, cultural, and governance readiness. Such holistic maturity ensures that data sharing is not an isolated technical act but a continuous, trusted, and value-oriented collaboration aligned with European Data Spaces principles of **Sovereignty**, **Interoperability**, **Decentralization**, **Transparency** and **Security**, and **Governance**.

In summary, the domain model illustrates how **Data Spaces function as federated ecosystems** where governance, trust, and interoperability converge. The vertical alignment of layers (business → information → trust → system → organization) shows that **sovereign data exchange** is not achieved by infrastructure alone but through a combination of **legal, semantic, and technical coordination** (European Commission, 2024; Otto et al., 2019; Curry & Tuikka, 2022). Each construct in the diagram is therefore an essential component of a trusted digital ecosystem that enables compliant, decentralized, and value-driven data sharing across sectors.

3.2 Data Space Architecture

Designing a Data Space requires not only defining its conceptual foundations but also establishing an architectural view that clarifies how these ideas materialize in practice. This section introduces the artefacts that operationalize the previously outlined principles, providing a blueprint for how a Data Space can be structured, governed, and technically supported. For this, two complementary models are proposed: a Conceptual Architecture, which organizes the core components and interactions, and a Technological Architecture, which specifies the enabling infrastructure and services required for implementation.

3.2.1 Data Space Conceptual Architecture

The goal of the conceptual architecture section is to define the standard components, responsibilities and interactions that constitute an inter-organizational Data Space, ensuring alignment with its core principles of trust, sovereignty, interoperability and federated governance. It provides a structured, more technology-agnostic model, that translates the domain concepts into architectural building blocks. These components will later be instantiated through the appropriate technologies during the implementation, forming the foundation upon which the enterprise Data Space will be built.

The proposed conceptual architecture, building upon the structure defined in the domain model and aligned with the International Data Spaces Reference Architecture Model (IDS-RAM), can be broken down into five major architectural blocks. Each block corresponds to a distinct set of responsibilities within the inter-organizational Data Space: (i) Federation and Metadata Catalogs, (ii) Sovereign Data Exchange, (iii) Policy-Driven Trust and Access Control, (iv) Domain-Level Data Product Platforms, and (v) the Process Layer.

Figure 3 presents the conceptual architecture and illustrates the relationships and interactions between these components. It was developed by the author as part of the proposed architectural contribution. It is an original representation that consolidates the main conceptual components identified in the literature and organizes them into a coherent architecture for inter-organizational Data Spaces.

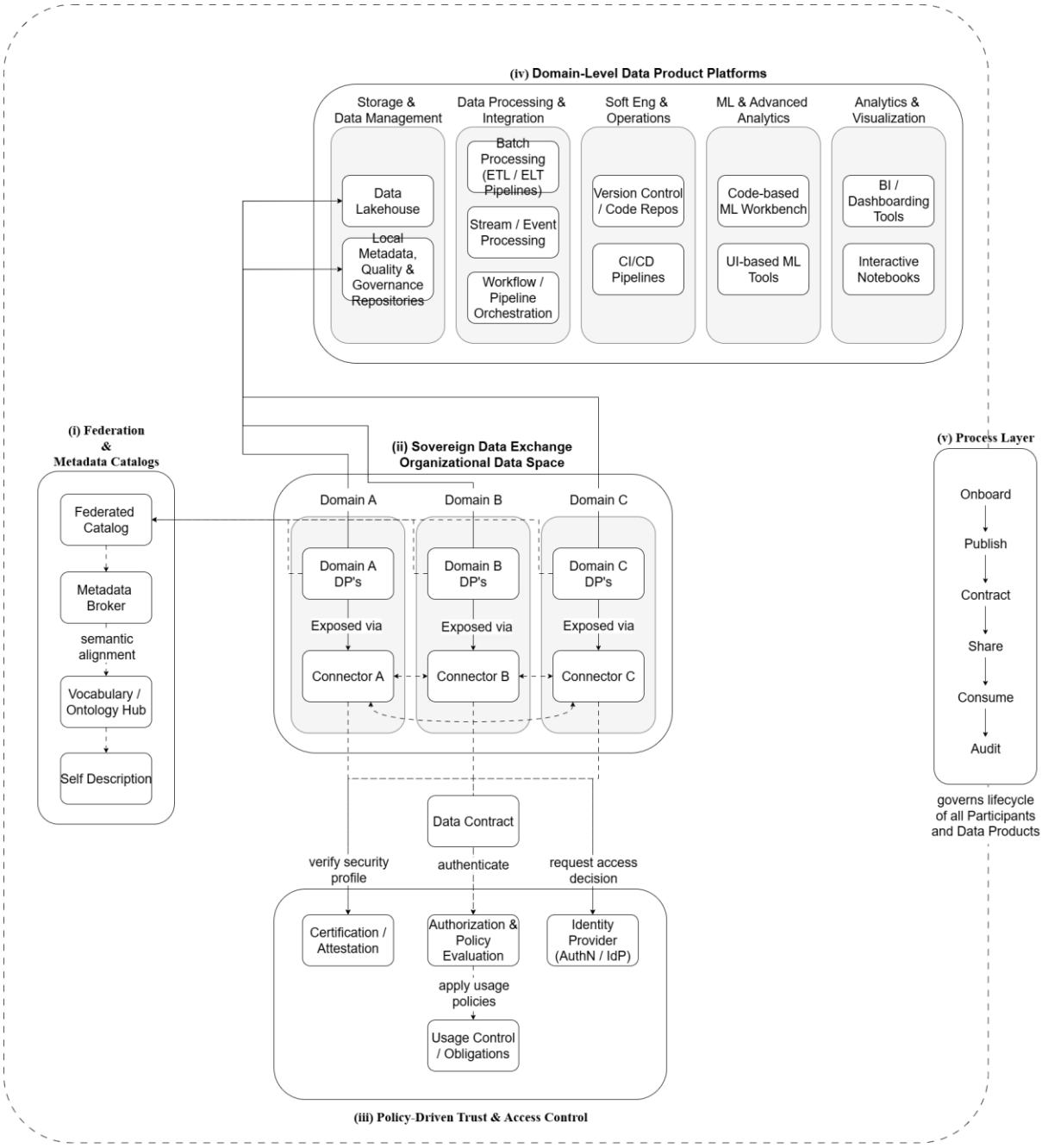


Figure 3: Data Spaces Conceptual Architecture

Inter-Domain Articulation

In the center of the proposed conceptual architecture lies the inter-organizational Data Space, represented as a federation of autonomous domains, each responsible for the creation and management of its own Data Products. Rather than enabling direct uncontrolled access between domains, the Data Space adopts the International Data Spaces principle that all cross-domain interactions must be mediated through certified Connectors. The IDS-RAM explicitly defines the Connector as the "central technological building block (…) allowing Participants to

exchange, share and process digital content", while ensuring that "the data sovereignty of the Data Owner is always guaranteed" (IDSA, 2019).

In this architecture, each domain exposes its Data Products exclusively through its Connector, which acts simultaneously as the technical access point and the enforcement boundary for identity, trust and usage policies. Connectors must be uniquely identifiable, authenticated, and capable of verifying the identity and security properties of other connectors before any data exchange takes place. As a result, interoperability across the Data Space is not implicit or based on technical integration alone, but intentionally governed through controlled, policy-aware, connector-to-connector interactions. This approach reflects the IDS emphasis on "standardized interoperability" among trusted peers, ensuring that data is exchanged directly between certified Connectors in a peer-to-peer model, without the need for centralized storage or intermediaries. While domains maintain full autonomy over their internal data management and technological choices, their participation in the Data Space requires adherence to a shared, federated trust and governance framework. This federated model preserves domain-level independence while enabling controlled collaboration across the organization, ensuring that data exchange occurs in a secure, verifiable and sovereign manner consistent with the core principles of the IDS-RAM.

Federation and Metadata Catalogs

A foundational element of the proposed Data Space architecture is the set of federated metadata services that collectively enable discovery, semantic interoperability, and the consistent interpretation of data assets across domains. In line with the IDS-RAM Information Layer, metadata is treated as an essential artefact: it is published, queried, versioned and interpreted across participants, while the underlying data remains sovereign and under the direct control of each domain. The IDS-RAM explicitly states that every data source participating in the ecosystem must be described on the Information Layer, enabling participants to describe, publish, maintain, and version metadata about their endpoints, schemas, vocabularies, and associated policies.

The Federated Catalog serves as the primary discovery mechanism, allowing users and domains to identify available Data Products without centralizing the underlying data. It indexes metadata published by domains through their Connectors and supports search, filtering, and classification based on descriptive and semantic attributes.

Complementing the catalog, the Metadata Broker acts as an intermediary component that receives, stores, and manages metadata supplied by participants. The IDS-RAM defines the Broker Service Provider as an entity that "stores and manages information about the data

sources available," providing structured query capabilities for Data Consumers and acting as a metadata registry for the ecosystem. Crucially, this brokering applies only to metadata: once metadata has been retrieved, the Broker plays no role in the subsequent data exchange. This reinforces the Data Space principle that metadata is federated, whereas data flows remain strictly peer-to-peer between connectors.

To ensure that metadata published by different domains is interpretable across heterogeneous domains, the architecture incorporates a Vocabulary / Ontology Hub. The IDS-RAM describes vocabulary hubs as servers that "store vocabularies, enable collaboration, and support activities such as search, selection, matching, updating, version management, and deletion," thereby enabling the creation of syntactic and semantic alignment across participants. At a deeper level, this builds on the IDS Ontology, the normative semantic representation of the IDS Information Model, which provides formal, machine-interpretable definitions of the entities, relationships, and constraints used to describe data assets and services. The presence of a shared vocabulary ensures that Data Product metadata is not only discoverable but also meaningfully comparable across domains, strengthening interoperability.

Finally, the Self-Description component plays a crucial role in exposing structured, machine-readable metadata about participants, connectors, and the resources they provide. It can be described as a metadata artifact containing identity information, interface descriptions, exposed data sources, associated policies, and contract-relevant attributes, enabling other participants to verify capabilities and trustworthiness before initiating exchange. Similarly, participant-level self-descriptions encode business, legal, and operational metadata, including roles, certifications, and domain classifications. Together, these descriptions allow the Data Space to establish a uniform understanding of who participates, what assets they expose, and under which conditions data can be accessed.

Taken together, these federated metadata services provide the informational backbone of the Data Space. They ensure that metadata, not data, is what circulates across the ecosystem; that semantics remain aligned across heterogeneous domains; and that participants and connectors can be reliably identified, interpreted, and assessed. This enables the inter-organizational Data Space to function as a coherent, interoperable environment despite the autonomy and technological diversity of its constituent domains.

Sovereign Data Exchange

The Sovereign Data Exchange block represents the controlled, contract-governed data exchange mechanism that occurs between organizational domains. In this model, each domain maintains full autonomy over its internal data management practices but exposes its Data

Products exclusively through a certified Connector. This design implements the IDS principle that all cross-domain interactions must occur via connectors. These act as the central technical component for enabling secure and sovereign data exchange and ensure that data access never bypasses the designated trust boundary of the Data Space.

Within the inter-organizational Data Space, domains do not communicate directly or retrieve data from one another as in traditional data integration models. Instead, connector-to-connector communication forms the only valid exchange pathway. Before the flow of any data, the IDS-RAM specifies that each Connector must perform mutual authentication, verify the integrity and identity of its counterpart, and establish a secure communication channel, ensuring that interoperability between domains is governed and intentional, rather than implicit or based on technical convenience. Actual access is granted only when the domains reach a **Data Contract** agreement.

A Data Contract is a machine-interpretable specification of the terms, conditions, and obligations that govern the consumption of a Data Product. In the IDS-RAM, this corresponds to the Contract Offer/Contract Agreement mechanism, where a provider issues a contract offer describing permitted purposes, usage restrictions, retention limits, obligations, and prohibitions. The consumer's connector evaluates the offer against its own policies and capabilities, and exchange can proceed only if both sides reach a mutually compatible contract agreement. In this sense, the Data Contract, rather than authentication alone, represents the decisive mechanism through which sovereign access is granted.

Once a contract has been agreed, connectors enforce the associated usage policies throughout the data flow. These policies accompany the data and are evaluated continuously, ensuring that operations such as read, store, compute, or onward share are performed only when explicitly allowed. This transforms data sharing into a policy-aware and contract-driven process: technical interoperability occurs exclusively under the conditions encoded in the Data Contract. During and after exchange, connectors generate audit logs capturing contract identifiers, permitted operations, access timestamps, and policy decisions. These audit trails provide accountability, allowing providers to verify that data was used in accordance with the agreed terms and enabling retrospective compliance checks.

Taken together, the Data Contract, connector-to-connector negotiation, and continuous policy enforcement constitute the core of sovereign exchange. Domains never share data directly; they share data only under a mutually accepted contract, enforced by certified connectors. This mechanism operationalizes the IDS vision of a federated, trust-based, and sovereignty-

preserving data sharing ecosystem, where technical, legal and organizational controls converge within a single contract-governed exchange model.

Policy-Driven Trust and Access Control

The Policy-Driven Trust and Access Control block provides the organizational mechanisms through which participants, connectors, and Data Products are authenticated, authorized, and governed within the Data Space. While the previous section focused on the mechanics of sovereign exchange, this block supplies the identities, credentials, policies, and security assurances required for those exchanges to occur in a secure and compliant manner. In IDS-RAM, these trust services form part of the System and Business layers and constitute the ecosystem's "trust anchors". They ensure that every interaction, whether it is metadata publication, contract negotiation, or data sharing, occurs only between authenticated and policy-compliant participants.

At the foundation of this block is the **Identity Provider (IdP)**, which issues and validates cryptographic identities for both participants and connectors, ensuring that every entity is uniquely identifiable and authenticated through a trusted authority. IDS-RAM requires connectors to rely on such trusted identity services rather than local identifiers, preventing impersonation and enabling verifiable trust relationships between domains.

Building on authentication, the **Authorization and Policy Evaluation** function determines whether a participant is permitted to access or use a Data Product, based on contractual terms, organizational roles, contextual attributes, and machine-interpretable usage policies. This component ensures that no Connector releases data before the access request has been validated against the conditions defined in the Data Contract and by the Data Provider.

The **Usage Control and Obligations** component extends sovereignty beyond initial access by enforcing how data may be processed after it has been retrieved. Whereas traditional access control governs *who* may access a resource, usage control governs *what happens after access,* enforcing retention limits, purpose restrictions, deletion obligations, or prohibitions on onward sharing throughout the runtime of data processing.

Finally, **Certification and Attestation** mechanisms verify the security posture and compliance level of participants and connectors. Certification bodies and standardized security profiles defined in IDS-RAM provide independently verifiable assurance that a connector satisfies required technical and organizational safeguards, establishing trust not through assumption but through formal assessment.

Together, these components form a federated trust fabric across the Data Space. Identity, authorization, usage control, and certification collectively ensure that every interaction,

whether metadata exchange, contract negotiation, or sovereign data sharing, occurs between authenticated, certified, and policy-compliant actors. This enables domains to remain autonomous while participating in a shared, trustworthy environment where all exchanges are governed by consistent, enforceable and sovereign rules.

Domain-Level Data Product Platforms

The Domain-Level Data Product Platforms represent the internal capabilities that each domain uses to design, build, manage, and maintain the Data Products that will later be exposed through its Connector. Whereas the federation and exchange layers govern cross-domain interoperability, this block concerns the local technical environment within each domain. It operationalizes the IDS-RAM principle that data remains under the control of its provider and that participants may rely on heterogeneous internal systems, provided that they expose standardized interfaces and metadata to the Data Space. Domains therefore retain full architectural autonomy internally while ensuring that the Data Products they publish conform to the shared semantics, governance rules, and interaction protocols of the federation.

The first capability group, **Storage & Data Management**, encompasses the domain's internal data platforms, such as Lakehouse or Warehouse systems, alongside metadata repositories and data quality controls. These components form the persistence layer for raw, processed, and curated data assets. Although not part of the Data Space itself, they constitute the foundation upon which Data Products are constructed. Local metadata and quality repositories ensure the maintenance of accurate schemas, lineage, and quality indicators, which are subsequently reflected in the metadata published to the Federated Catalog. Only a subset of this internal metadata, namely, the Data Product–level descriptions, is published to the Federated Catalog; all other operational metadata remains internal to the domain.

The **Data Processing and Integration** capabilities include the pipelines, transformation engines, and orchestration tools used to ingest, process, and prepare data. Domains may employ both batch and streaming technologies depending on source system characteristics. Workflow orchestration ensures that transformations are reliable, executable, and reproducible. These processes support the creation of high-quality, versioned Data Products, even though the Data Space remains agnostic to the specific technologies deployed internally.

The **Software Engineering and Operations** capabilities support the lifecycle management of Data Products. Version control systems, CI/CD pipelines, testing frameworks, and documentation tooling provide mechanisms for maintaining code quality, traceability, and deployment consistency. IDS-RAM does not prescribe specific engineering approaches, yet it

emphasizes that participants must ensure stable and secure system behavior when interacting with the Data Space, making strong engineering practices essential.

The **ML and Advanced Analytics** capabilities support the development and deployment of analytical models where relevant to domain needs. These may include both code-based workbenches and low-code tooling. While machine learning outputs can themselves be published as Data Products, the internal modelling processes remain opaque to the Data Space, maintaining the principle of local autonomy.

Finally, the **Analytics and Visualization** layer provides dashboarding tools, notebook environments, and other interfaces that allow domain users to explore, validate, and interpret data. These capabilities enable both technical teams and business users to assess Data Product quality and relevance before publication and during ongoing operation.

Taken together, the Domain-Level Data Product Platforms form the **local operational substrate** of the Data Space. They allow each domain to manage its own technologies, processes, and analytical workloads while producing Data Products that meet the semantic, governance, and interoperability requirements of the federation. This separation between internal autonomy and external standardization enables the Data Space to accommodate inter-organizational diversity while maintaining coherent inter-domain interaction.

Process Layer

The **Process Layer** governs the lifecycle through which participants and Data Products operate within the Data Space. It provides a federated procedural model that ensures all domains engage in a consistent, policy-aligned, and auditable manner. According to IDS-RAM, processes play a central role in coordinating how Participants and Connectors are onboarded, how they interact, and how they eventually disengage from the ecosystem, ensuring that data flows are embedded within a structured, accountable, and governance-driven framework.

The lifecycle begins with **participant onboarding**, during which a domain registers with the federation services, publishes its self-description, and obtains the credentials and certifications required for participation. This step establishes the identity, trust attributes, and technical capabilities of each participant. Once onboarded, domains may proceed to **Data Product publication**, where metadata describing the Data Product, its schema, semantics, policies, and endpoints, is exposed to the Federated Catalog via the Connector. This makes resources discoverable across the Data Space without replicating or centralizing the underlying data.

Once Data Products are published, Participants may enter into **contract negotiation**, a step that determines the terms under which data can be accessed and used. In the IDS model, this corresponds to the Contract Offer/Agreement mechanism, which formalizes the obligations,

permissions, and restrictions applied to a resource. Only once contractual conditions are validated can **sovereign data sharing** occur, always mediated through connector-to-connector interactions that enforce authentication, authorization, and usage policies.

Throughout the interaction, the Data Space supports **monitoring and auditing**, enabling providers to verify that data usage complies with contractual terms and organizational policies. Logging, auditing, and verification ensure accountability while supporting trust among participants.

Together, these steps: onboard → publish → contract → share → consume → audit — form a federated lifecycle that spans the entire architecture and ensures consistent, compliant, and policy-aware operation across all domains.

Taken as a whole, the five architectural blocks operate as an integrated framework for implementing an inter-organizational Data Space. The Federation and Metadata Catalogs establish trusted discovery and shared semantics; the Policy-Driven Trust and Access Control block provides the identity, authorization, certification, and usage-control foundations required for secure participation; and the Sovereign Data Exchange block enforces these guarantees by ensuring that all cross-domain interactions occur through certified, contract-governed connectors. The Domain-Level Data Product Platforms preserve internal autonomy while enabling the publication of interoperable Data Products, and the Process Layer binds all components together through a unified governance lifecycle.

By aligning these blocks with the conceptual model defined in Section 3.1, the proposed architecture forms a coherent bridge to the technological instantiation presented in Section 3.2.2. It provides a standards-aligned foundation for deploying a trusted, semantically interoperable, and sovereignty-preserving Data Space, while accommodating the autonomy and heterogeneous requirements of its constituent domains.

3.2.2 Data Space Technological Architecture

The conceptual architecture presented in the previous section establishes the high-level components, responsibilities, and interactions required for an inter-organizational Data Space. To translate this conceptual model into a deployable solution, it must be complemented with a technological architecture that identifies concrete platforms, software components, and infrastructure choices capable of implementing each architectural block. Figure 4 represents the Data Space Technological Architecture and illustrates how a representative set of technologies, with them being open source, commercially available and/or cloud-native, may be mapped to

the conceptual components. The goal is not to prescribe a single implementation, but rather to demonstrate how the principles and capabilities defined conceptually can be realized through widely available technologies, adaptable to different organizational contexts and maturity levels.

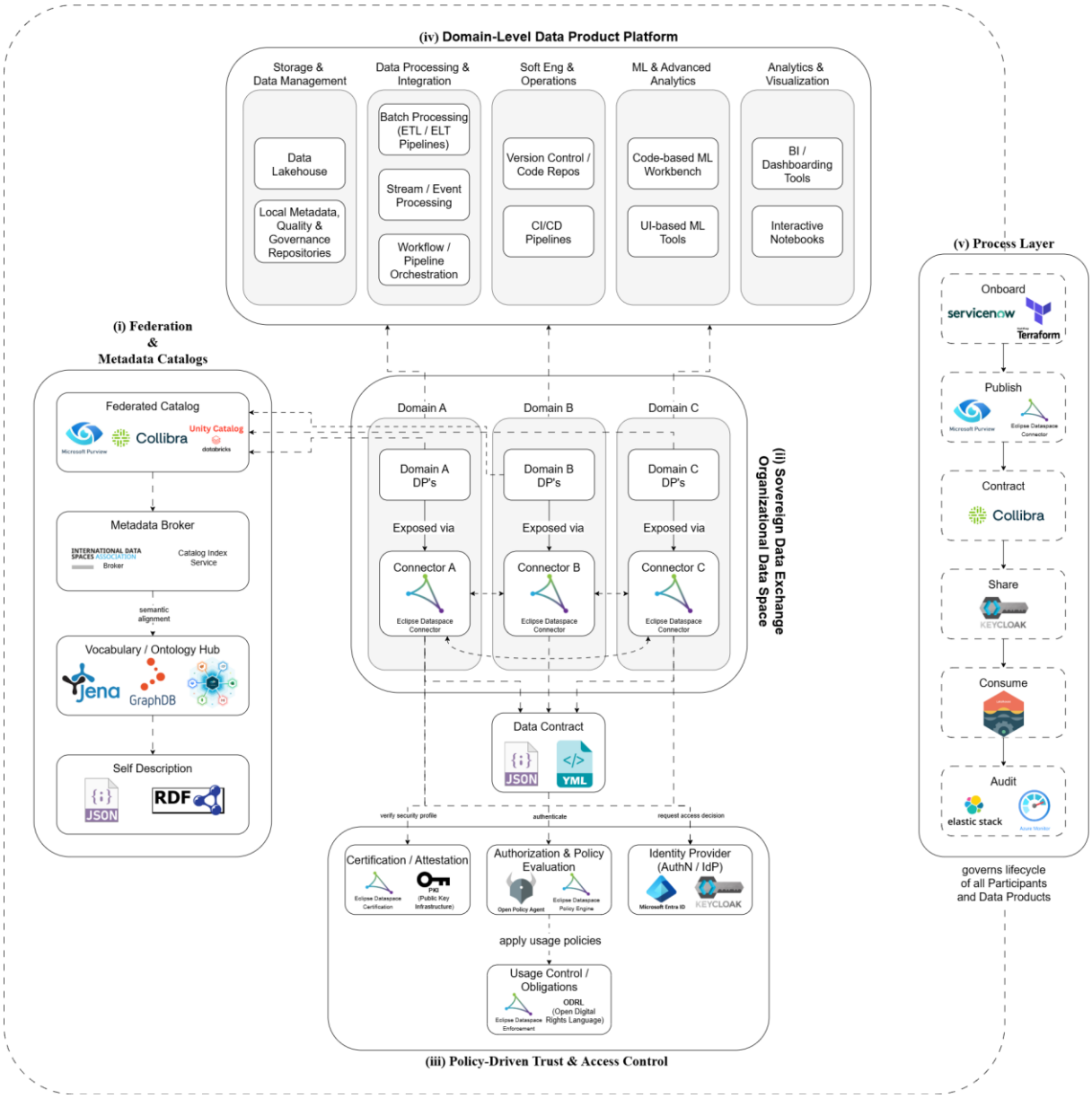


Figure 4: Data Spaces Technological Architecture

The Data Spaces Technological Architecture presented in Figure 4 was developed by the author as a derivation of the conceptual architecture introduced in Figure 3. It represents an original contribution that translates the conceptual model into a technological perspective, while being in accordance to IDS Architectural principles.

The first technological block, **Federation and Metadata Catalogs**, comprises the mechanisms that enable domains to publish, discover, and interpret Data Products in a consistent and machine-readable manner, in accordance with the IDS-RAM Information Layer. A Federated Catalog may be implemented through enterprise metadata platforms such as Microsoft Purview, Collibra, or Unity Catalog, which provide capabilities for metadata harvesting, search, lineage visualization, classification, and governance integration. These platforms allow domains to expose Data Product metadata without centralizing the underlying data, thereby preserving data sovereignty while enabling organization-wide visibility.

Within this federated metadata layer, the Metadata Broker plays a complementary role by acting as an intermediary service responsible for indexing, registering, and making Data Product metadata discoverable across the federation. While often implemented as part of a broader metadata platform, the broker conceptually aligns with the IDS Broker Service, enabling controlled publication and lookup of metadata without participating in data exchange or enforcing runtime policies.

Semantic interoperability across domains is supported through ontology and vocabulary services built on semantic web technologies, such as Apache Jena Fuseki, GraphDB, or Stardog. These components manage shared vocabularies, taxonomies, and conceptual models that define domain concepts, relationships, and governance terms in a machine-interpretable form. By establishing a common semantic layer, they ensure that Data Products published by heterogeneous domains can be interpreted consistently across the federation.

The **Self-Description Service** complements these capabilities by exposing structured metadata about participants, connectors, and the Data Products they offer. Self-descriptions include identity attributes, technical capabilities, governance declarations, and policy-related information, typically expressed using IDS-compliant models or semantic formats such as JSON-LD or RDF, which allow metadata to be represented in a structured, machine-interpretable form while remaining compatible with linked-data and semantic-web standards. Importantly, self-descriptions serve a purely informational role: they support automated discovery, validation, and trust establishment across the federation but are not involved in the runtime exchange of data.

The second technological block, **Sovereign Data Exchange**, is implemented through IDS-compliant connectors that operationalize secure, contract-governed, peer-to-peer communication between domains. A widely adopted implementation is the open-source Eclipse

Dataspace Components (EDC) Connector, which supports mutual authentication, encrypted communication channels, automated contract negotiation, and runtime policy enforcement. From a technological perspective, the connector acts as a controlled gateway between the domain's internal Data Product platform and the Data Space, decoupling internal storage and processing technologies from external data sharing concerns.

Policy and contract evaluation during exchange can be supported using technologies such as the EDC Policy Engine, Open Policy Agent (OPA), or organizational attribute-based access control mechanisms (e.g., Azure RBAC). These engines determine whether access requests meet the conditions defined in the Data Contract and enforce the obligations and restrictions associated with the Data Product.

Logging, monitoring, and audit capabilities ensure that data usage conforms to contractual and policy constraints. Platforms such as Azure Monitor, Azure Sentinel, Elastic Stack, and Splunk provide traceability of access events, policy decisions, and contract identifiers, enabling retrospective verification and supporting accountability across domains.

One essential component of this interaction are **Data Contracts**, a formal agreement that defines the technical, semantic, and governance conditions under which a Data Product may be accessed or consumed. In practice, they specify elements such as the agreed schema, data quality expectations, availability and freshness guarantees, permitted usage purposes, and policy or compliance constraints. These contractual specifications can be managed and governed using enterprise data governance platforms such as **Microsoft Purview** or **Collibra**, which allow organizations to document Data Contracts, manage approval workflows, version contract terms, and associate policy rules with specific Data Products. These contractual specifications are then enforced at runtime by the domain's connector, ensuring that contracts are not purely documentation artifacts but operational mechanisms governing access.

The third technological block, **Policy-Driven Trust and Access Control**, provides the identity, authorization, usage control, and certification foundations required for secure participation in the Data Space. Identity and authentication services may be implemented using Microsoft Entra ID, Keycloak, or Auth0, all of which provide trusted credential management and token-based authentication compliant with modern standards such as OAuth2 and OpenID Connect. These services ensure that connectors and participants are uniquely identifiable and interact only through trusted identities.

Authorization and policy evaluation, determining what a participant is permitted to access or do, can be enforced through Azure RBAC/ABAC, Open Policy Agent (OPA), or authorization

components embedded within EDC. These mechanisms ensure that no access is granted unless the request complies with both organizational roles and the terms of the relevant Data Contract. Usage control, which governs how data may be used after access has been granted, is enforced at runtime by the connector through embedded policy enforcement mechanisms. These mechanisms rely on machine-interpretable policy expressions, most commonly defined using the W3C Open Digital Rights Language (ODRL), which is well aligned with Data Space requirements for expressing data usage constraints and obligations. While alternative access control languages such as XACML may be used in enterprise environments, ODRL provides greater expressiveness for data-centric usage policies. Certification and attestation rely on IDS-recognized certification schemes or cloud security frameworks, providing formal assurance that connectors and participants meet the required security and governance criteria.

The fourth block, **Domain-Level Data Product Platforms**, are supported by the internal technological capabilities of each domain, which remain autonomous regarding infrastructure and tooling choices. Domains may employ a variety of technologies depending on organizational maturity, performance requirements, and existing systems.

Storage and data management capabilities can be built using Databricks Lakehouse, Azure Data Lake Storage, Delta Lake, Snowflake, or relational databases such as PostgreSQL. Metadata and quality controls may be supported through Unity Catalog, Microsoft Purview, Great Expectations, or Monte Carlo.

Data processing and integration pipelines can be implemented using Apache Spark, Databricks, Apache Flink, Azure Data Factory, or Azure Synapse Pipelines, supporting both batch and streaming ingestion patterns.

Software engineering and DevOps activities are supported by tools such as GitHub, Azure DevOps, GitLab, or Jenkins, ensuring reliable versioning, testing, documentation, and deployment of Data Products.

Machine learning, analytics, and visualization capabilities may rely on MLflow, Azure Machine Learning, TensorFlow, Power BI, or notebook environments. These tools enable domain teams to design analytical assets that can later be exposed as Data Products through the domain's connector.

These technologies allow each domain to manage its own workflows and platforms while still producing Data Products compatible with the semantic, policy, and interoperability requirements of the Data Space.

Finally, the **Process Layer** is supported by technologies that coordinate lifecycle activities such as onboarding, publication, contracting, monitoring, and auditing. Workflow and governance processes can be orchestrated using Collibra Workflow Engine, ServiceNow, or Azure DevOps Pipelines, ensuring that onboarding, certification, publication, and review cycles occur in a structured and auditable manner.

Infrastructure provisioning and environment standardization may additionally be supported through Infrastructure-as-Code tools such as Terraform, enabling reproducible deployment of connectors, metadata services, and supporting infrastructure across domains.

Operational monitoring and auditing activities can be supported through Azure Monitor, Grafana, Splunk, or audit services embedded in EDC, which provide visibility into data flows, contract enforcement, and the operational health of the Data Space.

Together, the technologies mapped to the five architectural blocks provide a coherent and implementable pathway for realizing the conceptual Data Space model. They demonstrate how open-source, commercial, and cloud-native tools can be combined to support federated discovery, semantic alignment, sovereign exchange, trusted participation, and domain-level Data Product creation. This technological architecture bridges the conceptual elements defined in Section 3.2.1 with the proof of concept implementation presented in the following chapter, enabling the deployment of a trusted, interoperable, and sovereignty-preserving inter-organizational Data Space while accommodating the autonomy and heterogeneity of its constituent domains.

4. DATA SPACES PROOF OF CONCEPT

Building on all the concepts already explored previously, this chapter presents a Proof of Concept (PoC) developed to validate the proposed approach for designing and implementing an inter-organizational Data Space. Building on the Domain Model, Conceptual and Technological Architectures defined in Chapter 3, the PoC serves as a simplified but representative environment where the fundamental behaviors of a Data Space can be demonstrated. Its purpose is not to replicate a full production scenario, but to illustrate how autonomous domains publish and consume from Data Products, how governance and interoperability mechanism operate in practice, and how data flows through the main architectural components. The chapter details the scope of the PoC, the organization of the participating domains, the infrastructure and tools used, and an example of a Data Product lifecycle, highlighting the essential elements required to materialize the Data Space in a controlled and comprehensible setting.

4.1 Scope of the Proof of Concept

This section defines the scope within which the PoC is developed. In this work, the PoC is just focused on demonstrating the essential behaviors required for sovereign data sharing, namely the publication of Data Products, metadata-driven discovery, cross-domain consumption, contract and grant-based access control. Thereby, the PoC is not applied to a concrete business or market domain, but instead, it prioritizes clarity and architectural validation rather than domain-specific completeness.

To illustrate these mechanisms, the PoC considers **three autonomous organizations**, each representing a participant of the Data Space. This choice reflects the fundamental principle that Data Spaces are inherently inter-organizational ecosystems, where independently governed entities share data according to common trust, governance and interoperability rules. The three organizations participating in the Data Space are:

- Organization A – Customer Services: Provides customer-related master data and descriptive attributes.
- Organization B – Operations and Service Delivery: Publishes operational and performance metrics.

- Organization C – Analytics and Insights Provider: Consumes data from the other two participants to produce analytical outputs.

These organizations maintain full sovereignty over their internal systems, governance processes, and Data Products. The Data Space acts as the trust and interoperability layer that enables secure, controlled, and traceable data exchange between them.
Each organization publishes one Data Product, resulting in the following structure:

1. Customer Profile (Organization A) – source-oriented Data Product containing customer-level information.
2. Operational Metrics (Organization B) – source-oriented Data Product describing activity and service indicators.
3. Engagement Insights (Organization C) – consumption-oriented Data Product generated through the combination of the first two.

While the first two Data Products originate directly from each organization's internal systems, the third is produced by Organization C through the governed use of data made available within the Data Space. This setup allows the PoC to illustrate how derived Data Products can be created by combining data from multiple autonomous participants under a federated governance model.
As previously mentioned, the emphasis is placed on demonstrating the flow of data and the interaction between Data Space participants, rather than on domain-specific optimization or operational complexity. Nevertheless, the inter-organizational nature of the PoC reflects the federated trust, governance, and sovereignty principles that distinguish Data Spaces from purely internal data architectures.
The Data Space PoC Scope and Architecture presented in Figure 5 was developed by the author as part of the proof of concept. It represents an original contribution and models a simplified inter-organizational scenario based on fictitious organizations created for demonstration purposes.

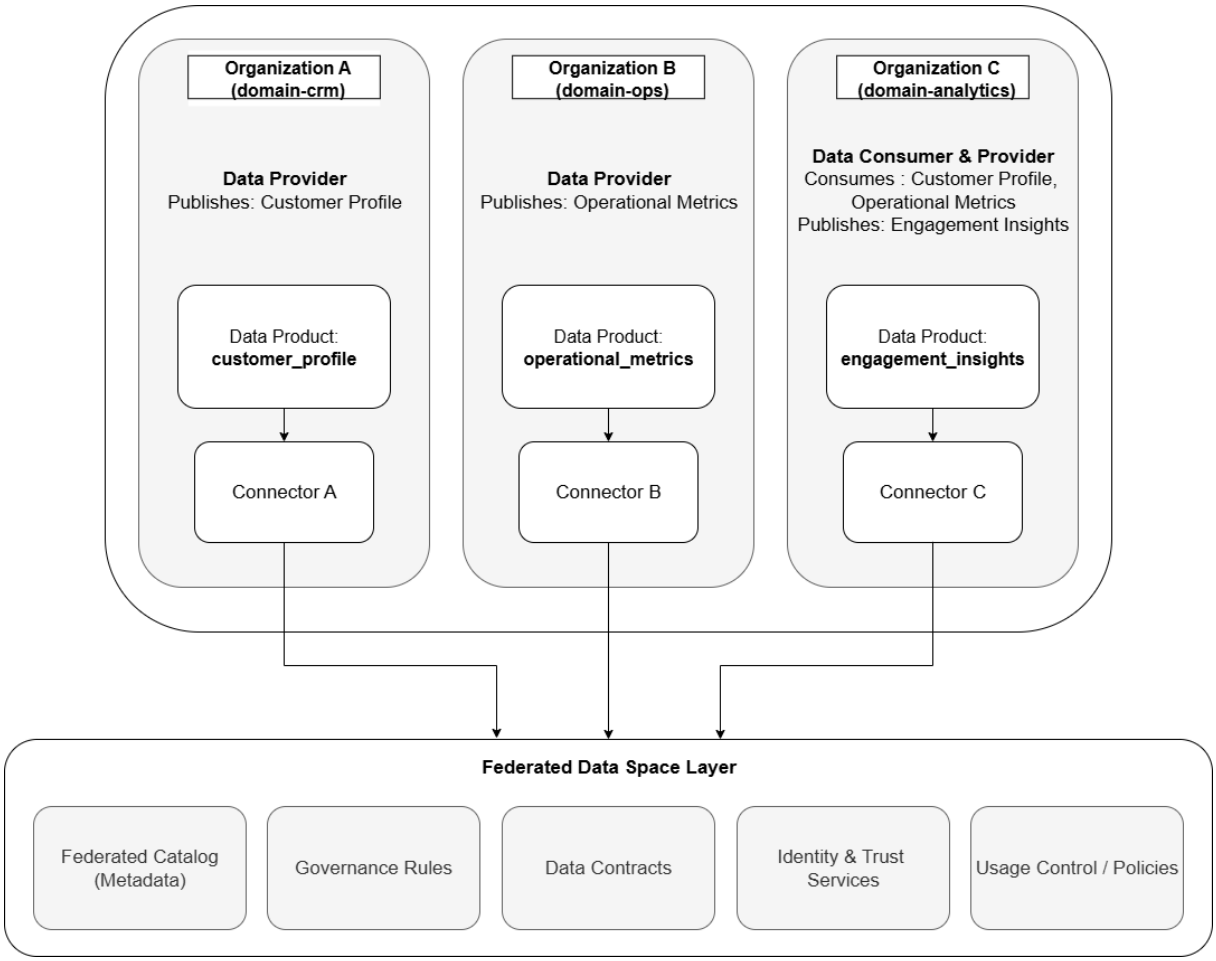


Figure 5: Data Space PoC Scope / Architecture

To ensure clarity in how data is exchanged within the PoC, each Data Product is defined using a simplified schema. These schemas reflect only the elements necessary to demonstrate publication, discovery, and consumption within the Data Space.

Table 3: Schema of Customer Profile Data Product (Organization A)

Attribute	Data Type	Description	Example
customer_id	string	Unique customer identifier within the Data Space context (domain-owned key).	CUST-102394
full_name	string	Customer full name (direct identifier; typically PII).	Ana Ribeiro
country_code	string	Customer country code (ISO 3166-1 alpha-2).	PT
subscription_type	string	Subscription category or plan type.	basic / standard / premium
customer_status	string	Customer lifecycle status.	active / suspended / churned
created_at	timestamp	Timestamp when the record was created in the source system.	2024-01-10T09:15:00Z
updated_at	timestamp	Timestamp of last update to the record.	2024-06-01T14:20:00Z
source_system	string	Identifier of the originating system.	crm_salesforce
record_version	integer	Record version for change tracking / optimistic concurrency / contract evolution.	2
pii_flag	boolean	Indicates whether the record contains personally identifiable information.	true

Table 4: Schema of Operational Metrics Data Product (Organization B)

Attribute	Data Type	Description	Example
customer_id	string	Customer identifier associated with support activity.	CUST-102394
metric_date	date	Business date the metrics refer to (daily granularity).	2025-12-01
activity_level	string	Qualitative activity label derived from operational volume.	low/medium/high
tickets_opened	integer	Number of tickets opened on the metric_date.	2
tickets_closed	integer	Number of tickets closed on the metric_date.	2
avg_resolution_time_hours	decimal(4,2)	Average resolution time for closed tickets (hours).	2.4
sla_breaches	integer	Count of SLA breaches observed on the metric_date.	0
updated_at	timestamp	Timestamp of last update to the metrics record.	2025-12-20T23:50:00Z
source_system	string	Identifier of the operational/support source system.	SAPHANA
data_quality_score	decimal(4,2)	Data quality indicator for this record (completeness/validity checks).	0.97

Table 5: Schema of Engagement Insights Data Product (Organization C)

Attribute	Data Type	Description	Example
customer_id	string	Customer identifier used to join insights back to domain data products.	CUST-102394
insight_date	date	Business date the insight refers to (aggregation/reference date).	2025-12-01
engagement_score	decimal(4,2)	Computed engagement indicator (scaled score).	87.4
churn_score	decimal(4,2)	Estimated churn likelihood (probability-like score).	0.18
score_method_version	string	Version of the scoring methodology/model used (governance + reproducibility).	v1.3.2
features_snapshot	json	Snapshot of key input features used to compute scores (minimal, auditable).	{"orders_30d":3, "logins_7d":5}
computed_at	timestamp	Timestamp when the insight was computed.	2025-12-02T03:10:00Z
input_dp_versions	json	Versions of upstream Data Products / datasets used as inputs.	{"CustomerProfile":"7", "CustomerSupportMetrics":"12"}
updated_at	timestamp	Timestamp of last update to the insight record (e.g., recompute).	2025-12-02T03:15:00Z

In summary, the PoC establishes a minimal but representative environment for validating the architectural and governance principles presented in Chapter 3. By modeling three autonomous organizations and their interactions through the Data Space, the PoC demonstrates how Data Products are published, discovered, accessed, and combined within a federated setting.

4.2 Proof Of Concept Design Principles

The proof of concept was designed with the objective of demonstrating the core principles of a Data Space in a clear and simplified manner. Rather than focusing on technical optimization or production-ready solutions, the design emphasizes simplicity, transparency, and alignment with the conceptual model presented in the previous chapters.

The focus of the PoC is the development of explicit and well-defined artifacts to represent the main elements of the Data Space. Participants, Data Products, contracts, access decisions, and catalog entries are all described through dedicated configuration files and infrastructure. This artifact-driven approach ensures that data ownership, governance rules, and sharing agreements are visible and easily traceable, reducing ambiguity and preventing uncontrolled or implicit data access.

Data sharing within the PoC is governed through **data contracts and approval processes**. Each Data Product is associated with a baseline contract defined by its owner, while data sharing with other participants requires the definition of an interaction contract and an explicit access approval. This design reflects the Data Space principle that data exchange is based on agreed conditions and clearly defined responsibilities, rather than on unrestricted technical connectivity.

To support traceability and reproducibility, all artifacts are managed using **version-controlled GitHub repositories**. This allows governance decisions, contract evolution, and Data Product changes to be tracked over time, providing a clear and auditable link between access requests, approvals, and the corresponding contractual agreements.

The PoC also follows a model of **decentralized ownership with federated coordination**. Each organization manages its own Data Products and related artifacts independently, while shared registries and catalogs provide a common layer for discovery and coordination across participants. This separation preserves organizational autonomy while enabling interoperability within the Data Space.

At the technical execution level, the PoC relies on an analytical environment developed in Databricks, used exclusively as a controlled execution and consumption layer. Within this environment, Data Products are exposed through final, consumption-ready views that are managed and governed by their respective domains, without introducing centralized ownership or authority over the data. These views represent the concrete access point for data consumers and correspond to the interfaces declared in the Data Product specifications and contracts. By

aligning contractual definitions with the exposed analytical views, the PoC ensures that data consumption remains consistent with the agreed sharing conditions.

Summarizing, this PoC prioritizes **process and governance clarity over a fully automated implementation**, as a means to validate the proposed Data Space approach. Access to data is considered valid only when an explicit request has been approved and recorded, and when a corresponding interaction contract exists. Although advanced enforcement mechanisms are out of scope, the complete governance workflow, from request submission, through approval, to authorized data consumption, is explicitly implemented and documented. This approach allows the PoC to demonstrate how Data Space governance operates in practice, without introducing unnecessary technical complexity.

4.3 Proof of Concept Implementation

This section describes the concrete implementation of the Proof of Concept, detailing how the Data Space concepts introduced in the previous chapters were materialized through a set of explicit artifacts and processes. The implementation follows the design principles outlined in Section 4.2 and focuses on demonstrating the full lifecycle of Data Products within a federated Data Space, from publication and governance to controlled consumption and derivation.

Rather than presenting isolated technical components, this section adopts a process-oriented perspective, reflecting how data sharing and governance operate in practice within a Data Space. Each subsection corresponds to a specific step in the implementation sequence.

4.3.1 Participant Definition and Trust Establishment

The first step of the implementation consisted in the explicit definition of the Data Space participants. All organizations participating in the PoC were registered in a shared participant registry. This registry was implemented as part of a dedicated and shared GitHub repository, named *dataspace-registry*, and contains a configuration file (*participants.yml*) under a *participants* folder. The file assigns a unique identifier to each organization and records basic descriptive information relevant for governance and coordination.

By relying on an explicit participant registry, the PoC ensures that all interactions within the Data Space are grounded on clearly identified and pre-registered entities. All Data Products, contracts, access requests, and governance decisions consistently reference these participant

identifiers, establishing a common and unambiguous notion of organizational identity across the PoC.

Participation in the Data Space is therefore restricted to organizations that have been explicitly registered in the participant registry. This registration acts as a prerequisite for publishing Data Products, requesting access, or approving data-sharing agreements. While advanced trust mechanisms such as digital certificates or cross-organizational identity federation are outside the scope of this work, the explicit definition of participants provides a clear and controlled boundary for interaction. This approach reflects the Data Space principle that trust and accountability are established through clearly defined roles, responsibilities, and governance processes, rather than being implicitly derived from technical connectivity alone.

Participant Registry

This artifact defines the set of Data Space participants by assigning a unique identifier to each organization and recording basic metadata, including domain affiliation, roles, contact points, and trust status. These identifiers are consistently referenced across all Data Products, contracts, and governance artifacts within the PoC.

The complete participant registry is provided in *Appendix 1 – Figure 6: PoC Participants: dataspace-registry\participants\participants.yml*

4.3.2 Data Product Publication

Once participants were defined, each organization published its Data Product within its own domain repository. This reflects decentralized ownership, with each organization fully responsible for the specification, documentation, and evolution of its Data Products.

Each Data Product is described using a dedicated specification file (*product.yml*), which captures essential information such as purpose, ownership, interface, and schema. This specification serves as the authoritative description of the Data Product within the Data Space and is referenced by other governance artifacts.

In addition to the specification, each Data Product is associated with a **baseline contract**, defined and maintained by the Data Product owner. The baseline contract expresses the default conditions under which the Data Product may be shared, including usage constraints, data sensitivity considerations, retention expectations, and high-level data quality requirements. By separating the product specification from the contractual terms, the implementation clearly

distinguishes between what the Data Product is and under which conditions it may be consumed.

This publication process reinforces ownership and accountability, as each organization explicitly defines both the content and the sharing conditions of its Data Products.

Data Product Specification

The Data Product specification describes the Data Product's identity, ownership, lifecycle status, exposed interface, and schema. It serves as the authoritative reference for the structure and interface of the Data Product and is used by contractual and catalog artifacts.

The full Data Product specification is included in *Appendix 2 – Figure 7: PoC Data Product Definition: domain-crm\customer_profile\product.yml*

4.3.3 Contract Definition and Data Sharing Agreements

To enable data sharing between organizations, the PoC introduces a clear distinction between **baseline contracts** and **interaction contracts**, each addressing a different level of governance within the Data Space.

Baseline contracts are defined and owned by the Data Product producer and represent the default conditions under which a Data Product may be shared within the Data Space. These contracts describe general usage rules that apply to all potential consumers, such as data classification, permitted purposes, retention expectations, and high-level data quality assumptions. A baseline contract does not grant access to any specific consumer; instead, it establishes the producer's initial position regarding how the Data Product can be used and under which constraints.

Baseline Contract

The baseline contract specifies the default sharing conditions of a Data Product, including allowed purposes, prohibited uses, retention rules, security assumptions, and data quality expectations. It defines governance constraints but does not grant access to any specific consumer.

A complete baseline contract is provided in *Appendix 3 – Figure 8: PoC Data Contract: customer_profile\contracts\baseline_contract.yml*

In contrast, **interaction contracts** are created to govern concrete data-sharing relationships between a specific producer and a specific consumer. Interaction contracts refine and specialize the baseline conditions by incorporating consumer-specific constraints, such as a reduced data scope, stricter usage limitations, or additional responsibilities. In the PoC, interaction contracts are defined to regulate the sharing of the *Customer Profile* and *Operational Metrics* Data Products with the Analytics organization, explicitly capturing the terms under which these datasets are consumed.

Interaction Contract

The interaction contract formalizes a concrete data-sharing agreement between a specific provider and consumer. It refines the baseline contract by defining consumer-specific scope, usage purpose, and governance constraints for the shared Data Product.

A representative interaction contract is included in *Appendix 4 – Figure 9: PoC Data Contract: operational_metrics\contracts\org-c-analytics-provider.yml*

Both baseline and interaction contracts are implemented as explicit artifacts and are stored alongside the corresponding Data Products within the producer's domain repository. This organization reinforces the principle that Data Product owners retain authority over the contractual terms governing their data, even when it is shared across organizational boundaries. By externalizing contracts as standalone and reviewable artifacts, the PoC avoids implicit or undocumented agreements and ensures that all data exchanges are governed by transparent and traceable conditions.

4.3.4 Governance Workflow and Access Grants

Within the PoC, governed data sharing is implemented through an explicit governance workflow that separates **the formal registration of access decisions** from **the process through which those decisions are made**. This separation ensures clarity between governance outcomes and governance procedures, while maintaining full traceability across the Data Space. Authorized data-sharing relationships are formally recorded through **access grants**, maintained in a shared governance registry (*access_grants.yml*). Each access grant represents the outcome of a governance decision and constitutes the authoritative record defining which participant is allowed to access a given Data Product and under which conditions. An access grant explicitly links the requesting participant, the providing participant, the Data Product being shared, and the interaction contract governing the exchange. By consolidating these elements into a single

artifact, access grants provide a clear authorization context and establish accountability for data consumption. Within the PoC, access to a Data Product is considered valid only when a corresponding access grant exists, reinforcing the principle that governance decisions are explicit, auditable, and traceable.

Access Grants Registry

This artifact records approved data-sharing authorizations by linking a requesting participant, a providing participant, a Data Product, and the governing interaction contract. It represents the authoritative record of access decisions within the PoC.

The full access grants registry is provided in *Appendix 5 – Figure 10: PoC Access Grants: dataspace-registry\governance\access_grants.yml*

The creation of an access grant is preceded by a structured approval process. Access requests are submitted using GitHub Issues in the relevant domain repositories, providing a lightweight but effective mechanism to document the requesting participant, the intended usage, and the proposed interaction contract.

Each request is evaluated by the **Data Product owner**, who is responsible for deciding whether the requested access complies with the baseline contract and with the producer's governance policies. While governance within the Data Space is federated and shared through common rules and registries, the authority to approve access to a specific Data Product remains local to its owner, preserving organizational sovereignty. Upon approval, the decision is formalized by registering the corresponding access grant in the shared registry, linking the approval process to its concrete governance outcome.

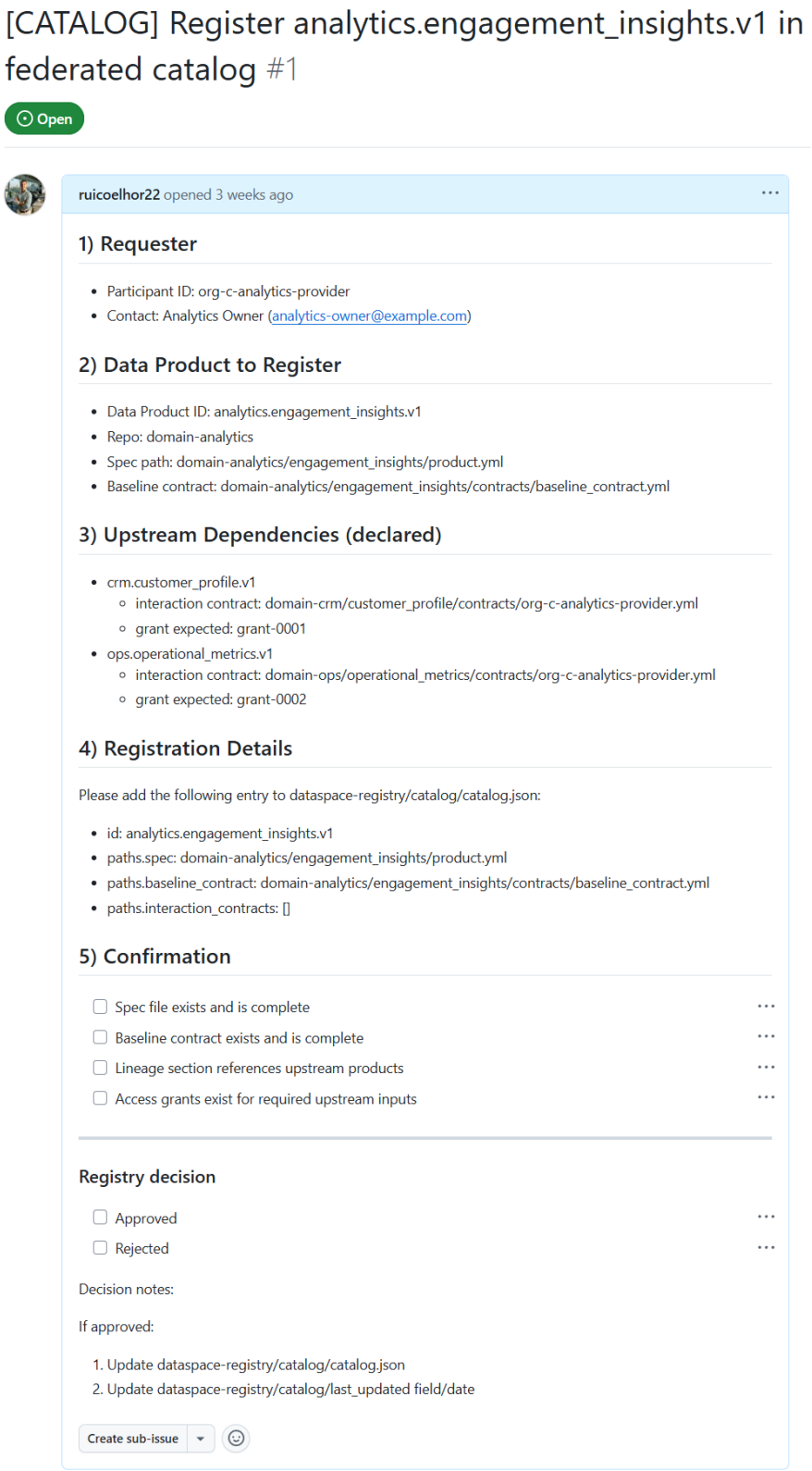


Figure 6: PoC Access Request – GitHub Issue

This workflow reflects a core Data Space principle: participation and coordination are governed at the Data Space level through shared processes and artifacts, while access to data is governed at the Data Product level by its owner. By explicitly modeling both aspects, the PoC

demonstrates how federated governance and local decision-making coexist within a controlled and transparent data-sharing ecosystem.

4.3.5 Federated Catalog Registration

To support discovery and coordination across the Data Space, all Data Products participating in the PoC are registered in a federated catalog, implemented as a shared metadata index (*catalog.json*). The catalog represents the onboarding and discovery layer of the Data Space, defining which Data Products are recognized as part of the ecosystem and can be discovered by other participants.

Federated Catalog

The federated catalog registers Data Products available within the Data Space by referencing their specifications, contracts, ownership, and exposed interfaces. It supports discovery without duplicating or centralizing Data Product definitions.

The complete catalog definition is included in *Appendix 6 - Figure 12: PoC Federated Catalog: dataspace-registry\catalog\catalog.json*

Each catalog entry references the corresponding Data Product specification and baseline contract, and, where applicable, the associated interaction contracts. Rather than duplicating metadata, the catalog acts as a registry of references, pointing to authoritative artifacts maintained within each domain repository. This design ensures that ownership and responsibility for Data Product definitions and contracts remain fully decentralized, while still enabling cross-organizational visibility.

Registration in the catalog constitutes the final step of Data Product onboarding into the Data Space. A Data Product becomes discoverable only after being registered, but such registration does not imply unrestricted access. Instead, the catalog provides the necessary information for potential consumers to understand the existence, ownership, and sharing conditions of a Data Product, serving as a prerequisite for initiating governed access requests.

Although technically centralized in its implementation, the catalog operates on a federated basis. It enforces common discovery and registration rules defined at the Data Space level, while preserving organizational sovereignty over data and access decisions. This separation between discovery, onboarding, and access control reflects a fundamental characteristic of Data Space architectures.

In a production-oriented scenario, and outside the scope of this PoC, the federated catalog could be implemented using a platform-native metadata service such as **Databricks Unity Catalog**. In such a setup, metadata management, lineage, and access policies could be natively enforced at the platform level, enabling tighter integration between Data Product specifications, governance rules, and consumption interfaces. While this level of enforcement could not be implemented in the present PoC due to platform permission constraints, the artifact-based catalog adopted here mirrors the conceptual role of such services, allowing the core principles of federated discovery and governance to be demonstrated independently of a specific technological implementation.

4.3.6 Analytics Data Product Implementation

The final step of the PoC implementation consists in the creation of the **Engagement Insights** Data Product by the Analytics organization. This Data Product represents the culmination of the proposed Data Space approach, demonstrating how governed data sharing across autonomous participants enables the creation of new, value-added Data Products.

The Engagement Insights Data Product is derived through the controlled consumption of the *Customer Profile* and *Operational Metrics* Data Products, strictly in accordance with the conditions defined in the approved interaction contracts. This ensures that data reuse does not bypass governance mechanisms, but instead operates within the contractual and organizational boundaries established earlier in the PoC.

To preserve traceability and accountability, the implementation explicitly declares **lineage relationships** to its upstream inputs and records metadata relevant to reproducibility and governance. This includes information such as the version of the scoring methodology used and references to the versions of the upstream Data Products consumed. By capturing this metadata as part of the Data Product definition, the PoC ensures that derived insights can be traced back not only to their source data, but also to the contractual agreements under which that data was accessed.

The Engagement Insights Data Product is exposed through **consumption-ready analytical views**, which constitute the concrete interface for downstream use. These views are aligned with the interfaces declared in the Data Product specification and reflect the contractual constraints agreed upon during the governance process. As a result, the output of the analytics process is itself treated as a first-class Data Product, subject to the same principles of ownership, discovery, and governed access as source-oriented products.

Through this final step, the PoC demonstrates that a Data Space is not limited to controlled data sharing, but also supports the **governed creation of derived Data Products**. While the PoC focuses on analytical aggregation and scoring, the same mechanisms could be extended to more advanced use cases, such as machine learning pipelines or the training of artificial intelligence models using data from multiple organizations. These scenarios would follow the same lifecycle - discovery, contractual agreement, governed access, and traceable derivation, but are considered outside the scope of this work in order to preserve focus and simplicity.

This implementation therefore illustrates the complete Data Product lifecycle envisioned by the proposed Data Space approach, from publication and discovery, through governed access, to controlled derivation and reuse, providing concrete validation of the model presented in the previous chapters.

4.3.7 Proof Of Concept Artifact Mapping

This section provides a consolidated overview of the main artifacts created in the Proof of Concept and their relationships. The purpose of this overview is to complement the implementation described in the previous sections with a clear mapping between Data Products, governance artifacts, and execution components.

The diagram below presents the PoC architecture, developed by the author as part of the proof of concept, representing an original contribution and a simplified inter-organizational scenario based on fictitious organizations created for demonstration purposes. It shows the participating organizations, their respective Data Products, and the supporting governance artifacts. Each organization maintains its own domain repository, where Data Products are specified and governed through Data Product specifications and contracts. This structure reflects decentralized ownership, ensuring that each organization remains responsible for its data and the conditions under which it is shared.

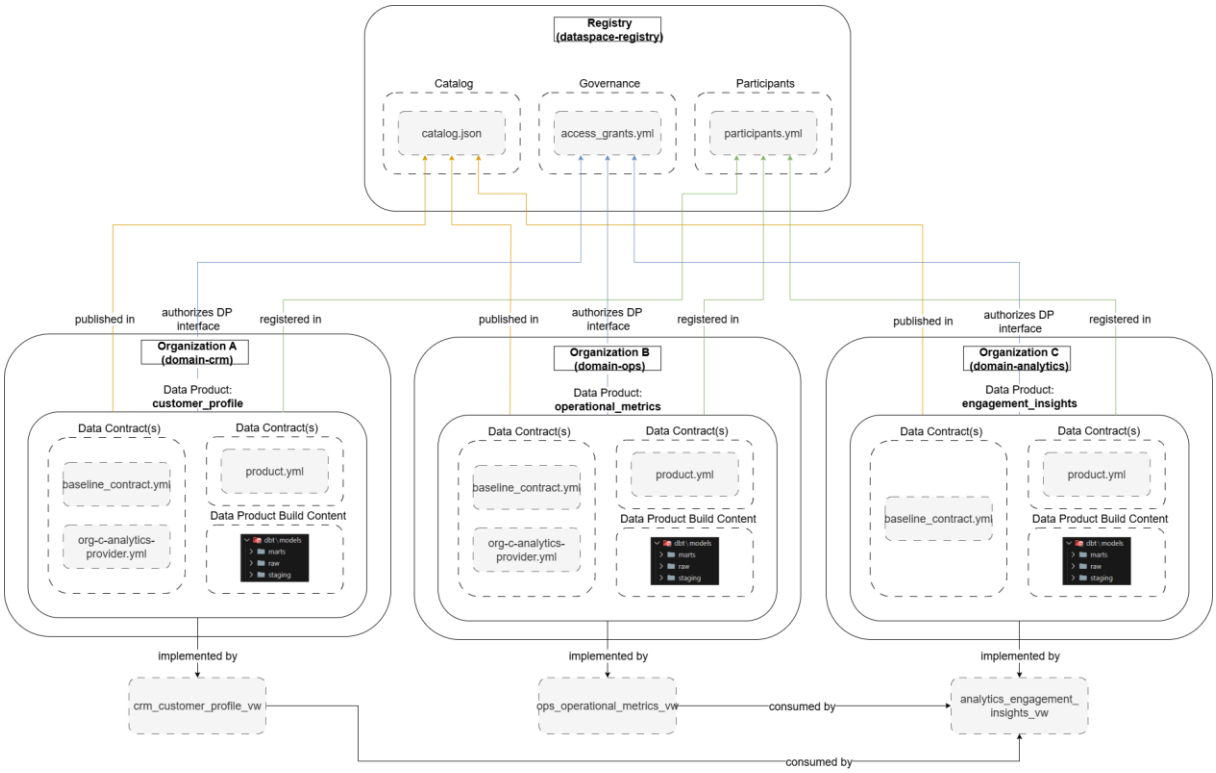


Figure 7: PoC Artifact Mapping

Shared governance artifacts, including the participant registry, access grants, and the federated catalog, are maintained separately in a common registry repository. These artifacts support onboarding, discovery, and access coordination, but do not store or process data. This separation makes explicit the distinction between data ownership and governance coordination within the Data Space.

At the execution level, Data Products are exposed through consumption-ready analytical views in a shared analytical environment. These views constitute the concrete access interfaces for data consumption and correspond to the interfaces declared in the Data Product specifications and contracts. Data exchange between organizations occurs only through these interfaces and only when supported by approved access grants.

In this PoC, connectors are not implemented as standalone technical components. Instead, their role is realized through the combination of Data Product interfaces, contracts, access grants, and catalog entries.

As mentioned, all PoC artifacts are maintained in their respective repositories, all managed under a dedicated GitHub organization, which serves as the centralized entry point for accessing, reviewing, and versioning the complete set of Data Space configuration files, contracts, and governance registries. It can be found in: https://github.com/dataspace-poc-org

Together, these artifacts enable controlled and traceable data exchange, allowing the core behaviors of a Data Space to be demonstrated without introducing additional technical complexity.

4.4 Validation of the Proposed Approach

In the following section, the validation of the proposed Data Space approach is performed through a qualitative assessment of the implemented Proof of Concept, focusing on the alignment between the conceptual principles discussed in the literature and the concrete artifacts and processes realized in practice. Rather than relying on performance or scalability metrics, the validation emphasizes architectural correctness, governance coherence, and traceability, which are central concerns in Data Space design and align with the purpose of the PoC.

According to the IDS-RAM (International Data Spaces Association, 2019), Data Spaces operate based on five core principles that ensure secure, interoperable, and sovereign data exchange.

The first principle validated by the PoC is **data sovereignty**. In the implemented approach, each participating organization retains full control over its Data Products, including their specifications, contractual conditions, and access decisions. Data Products are developed and maintained within domain-specific repositories, and access to them is granted exclusively by their respective owners. This confirms that the proposed approach enables data sharing without requiring organizations to resign ownership or control over their data assets.

The principle of **interoperability** is validated through the use of standardized Data Product specifications, structured schemas, and well-defined consumption interfaces. Metadata, contracts, and lineage information are represented in consistent and machine-readable formats, enabling Data Products to be discovered, interpreted, and consumed across organizational and technological boundaries. As a result, participants are able to collaborate, share and consume data in a consistent manner, without requiring prior bilateral technical alignment or bespoke integrations.

A **decentralized approach** is also explicitly realized in the PoC. Data remains physically managed within each organization's environment and domain repository, and is only consumed by other participants under approved conditions. There is no central data storage or consolidation of datasets within the Data Space. No centralized ownership or authoritative repository of data is introduced; data access remains controlled by providers and governed by

explicit contracts and grants. This ensures that ownership and responsibility remain with the data providers, preserving security and autonomy while still enabling controlled data exchange. The PoC further validates the principle of **transparency and security** through explicit mechanisms for identity definition, contractual agreements, and traceable access decisions. Participants are registered in a shared registry, data-sharing conditions are formalized through baseline and interaction contracts, and access approvals are explicitly documented through governance artifacts. Together, these mechanisms ensure transparency, accountability, and compliance with agreed rules, establishing trust and security through governance rather than technical assumptions.

Finally, the principle of **federated governance** is validated by the clear separation between shared coordination mechanisms and local decision-making authority. While onboarding, discovery, and governance transparency are supported through shared registries and catalogs, decisions regarding access to specific Data Products remain under the control of their respective owners. This distribution of responsibilities confirms that governance within the Data Space is collaborative and federated, rather than imposed by a central authority.

In addition to the validation of the individual principles, validation is further supported by the internal consistency of the implemented artifacts. Participant identifiers, Data Product specifications, contracts, access grants, and catalog entries are consistently referenced across repositories, ensuring coherence between governance decisions, discovery mechanisms, and data consumption. This consistency provides additional evidence that the proposed approach can be operationalized in a structured and reproducible manner.

Overall, the Proof of Concept demonstrates that the proposed Data Space approach successfully operationalizes the five core principles defined by the IDSA in a simplified but representative environment. While aspects such as automated enforcement, scalability, and advanced analytics are intentionally outside the scope of this work, the validation confirms the suitability of the approach as a foundation for secure, interoperable, and sovereign inter-organizational data ecosystems. The implications of these limitations, together with potential extensions, are discussed in the following chapter.

5. FINAL CONCLUSIONS AND FUTURE WORK

The purpose of this final chapter is to present the main conclusions derived from the development of this master's dissertation and to outline the main directions for future work. The chapter first synthesizes the key insights obtained from the literature review and state of the art, integrating them into the overall conclusions of the dissertation, and then discusses the conclusions drawn from the proposed approach and its validation through the proof of concept. Finally, it presents the main limitations of the work and identifies opportunities to expand, refine, or operationalize the contributions of this dissertation in both research and practice.

The literature review showed that Data Spaces emerge as a response to the limitations of previous data management paradigms, including Data Warehouses, Big Data Warehouses, Data Lakes, and Data Lakehouse models. Although these paradigms introduced relevant advances in scalability, flexibility, heterogeneity management, and analytical performance, they were not primarily designed to address controlled data sharing across organizational boundaries. As organizations increasingly collaborate with external partners, suppliers, customers, and institutions, purely intra-organizational architectures become insufficient to address requirements such as interoperability, data sovereignty, trust, semantic alignment, policy enforcement, and distributed governance.

The analysis also demonstrated that Data Spaces should not be understood solely as a technical architecture, but as a socio-technical paradigm that combines technological components, governance mechanisms, organizational roles, and shared rules for data exchange. Their relevance is associated with the need to enable autonomous participants to collaborate and reuse data while preserving control over their own data assets. In this sense, Data Spaces provide a structured foundation for trusted, interoperable, and sovereign data sharing in inter-organizational environments.

At the same time, the reviewed literature revealed that the field is still maturing. While existing principles, regulatory developments, and reference models provide important conceptual foundations, practical guidance for designing, deploying, and operationalizing Data Spaces remains limited. In particular, open questions persist regarding how conceptual, governance, and technological elements can be combined into coherent architectures, and how existing data platforms can be integrated into federated environments without compromising sovereignty, governance, or control. This gap justifies the contribution of this dissertation, which proposes a structured architectural approach and validates it through a proof of concept.

The remainder of this chapter is organized into two sections. Section 5.1 presents the conclusions regarding the proposed approach and the proof of concept, assessing the extent to which the developed artifact materializes the architectural principles explored throughout the dissertation and demonstrates the feasibility of implementing a federated inter-organizational Data Space. Section 5.2 discusses the main limitations of the work and outlines opportunities for future research and practical development.

5.1 Conclusions of the proposed approach

Throughout Chapters 3 and 4, the main artifacts defining the proposed Data Space approach were presented, namely the domain model, the conceptual architecture, and the technological architecture. The domain model synthesizes the key conceptual elements required to design and operate a Data Space, encompassing both technical components (such as Data Products, metadata services, and access mechanisms) and organizational aspects (such as participant roles, responsibilities, and governance processes). This dual perspective reflects the fundamental nature of Data Spaces as a paradigm that goes beyond infrastructure, addressing organizational structures, collaboration models, and shared governance principles.

The domain model serves as the foundation for the remaining artifacts developed in Chapter 3. Building upon this model, the conceptual architecture defines, at a high level, the components and interactions required to enable secure, interoperable, and sovereign data exchange across organizational boundaries. This architecture makes explicit the roles of Data Products, contracts, registries, catalogs, and governance mechanisms within a Data Space, while remaining independent of specific technologies.

The technological architecture further refines this perspective by illustrating how the conceptual components may be realized using concrete platforms and tools. Rather than prescribing a fixed implementation, this architecture provides illustrative examples of technologies that can support the proposed approach. This demonstrates that, in many cases, organizations may adopt a Data Space by reusing existing infrastructures and tools, provided that they reorganize their processes, governance models, and data management practices accordingly. The selection of technologies is therefore intentionally left open, allowing organizations to choose solutions that best fit their context and constraints.

Once these conceptual and architectural artifacts were consolidated, a Proof of Concept was developed to validate the assumptions underlying the proposed approach. The PoC

demonstrates that the proposed Data Space model can be instantiated as a functional, decentralized ecosystem, in which autonomous organizations publish, discover, and consume Data Products under explicit governance conditions. Although the PoC represents only a partial realization of the proposed approach, its implementation covers the essential elements required to validate the model, including participant definition, contractual data sharing, federated discovery, and governed Data Product derivation.

The results of the PoC confirm that the proposed approach enables decentralized data sharing while preserving data sovereignty and organizational autonomy. Data Products are treated as first-class entities, accompanied by explicit specifications, contracts, and governance artifacts. Discovery mechanisms allow participants to identify available Data Products together with their ownership, conditions of use, and supporting metadata, promoting informed and accountable data consumption. Importantly, the PoC shows that data sharing and reuse can occur without centralizing data or decision-making authority, relying instead on federated coordination and explicit agreements.

Beyond validating individual architectural components, the combined development of the domain model, conceptual architecture, and Proof of Concept demonstrates that Data Spaces can be approached as an artifact-driven and implementation-oriented paradigm. Rather than remaining at the level of abstract principles, the proposed approach translates governance, sovereignty, and interoperability requirements into concrete, versioned artifacts that can be independently developed, reviewed, and evolved by participating organizations. This contributes to bridging the gap identified in the literature between high-level reference models and actionable implementation guidance.

It is therefore concluded that this dissertation fulfills its main objective of proposing and validating a conceptual approach for the design and implementation of Data Spaces. By bridging theoretical foundations with a concrete, artifact-driven Proof of Concept, the work demonstrates how Data Space principles can be translated into practical models that support secure, interoperable, and sovereign data ecosystems.

5.2 Limitations and Future work

This section discusses the main limitations of the proposed Data Space approach, as evidenced through its partial realization in the Proof of Concept, and outlines directions for future work that directly stem from these limitations. The identified limitations do not reflect shortcomings of the conceptual model itself, but rather the deliberate simplifications and scope constraints

adopted during the PoC implementation. As such, they also represent natural opportunities for extending and strengthening the proposed approach in future research and practical applications.

A first limitation concerns the degree of technical enforcement of governance mechanisms. While the proposed approach assumes the existence of automated policy enforcement, identity management, and access control capabilities, the PoC implements these aspects primarily through explicit artifacts and documented processes. This choice was intentional, allowing the conceptual validity of the approach to be demonstrated without introducing platform-dependent complexity.

Future work may therefore focus on the implementation of fully automated governance enforcement mechanisms, including cross-organizational identity federation, policy evaluation engines, and fine-grained access control. Integrating platform-level governance services would allow contractual conditions and usage restrictions to be enforced dynamically and at scale, strengthening the operational maturity of the approach.

Another limitation relates to operational scalability and performance. Although the proposed approach is intended to support large, heterogeneous Data Space ecosystems, the PoC does not evaluate performance, scalability, or resilience under realistic production conditions. These aspects remain inherent to the approach but are not empirically validated within the scope of this work.

Future research could address this limitation by deploying the proposed model in larger-scale environments, involving a higher number of participants and Data Products, and by assessing system behavior under increased data volumes, concurrent access, and evolving governance conditions.

The trust and identity mechanisms are also simplified in the PoC when compared to those envisioned by the proposed approach. While participant identity and governance responsibilities are explicitly defined, advanced mechanisms such as certification frameworks, trust anchors, or federated identity infrastructures are not implemented. These elements are recognized as essential components of production-grade Data Spaces and are assumed as part of the broader conceptual approach.

Accordingly, future work could explore the integration of standardized trust frameworks and identity management solutions, enabling stronger assurances regarding participant authenticity, compliance, and accountability within the Data Space.

Another identified limitation concerns semantic interoperability. The PoC does not demonstrate semantic alignment in depth (e.g., ontology management or shared vocabularies), as schemas

were intentionally simplified and harmonized upfront to preserve clarity. While sufficient for illustrative purposes, this approach does not reflect the complexity typically encountered in real-world Data Space ecosystems.

Future work could therefore investigate mechanisms to support semantic interoperability at scale, such as shared domain ontologies, semantic mediation services, or tooling to assist schema evolution and cross-domain data understanding, reducing integration effort while preserving organizational autonomy.

The organizational and socio-technical dimensions of Data Spaces also remain only partially explored in this work. Beyond technical architectures, the successful adoption of Data Spaces requires organizational alignment, clearly defined roles, and cultural shifts toward data sharing and accountability.

Future studies could analyze how different governance structures, incentive models, and organizational arrangements influence the effectiveness and sustainability of Data Space initiatives, complementing the technical perspective adopted in this dissertation.

Additionally, future work could extend the proposed approach to support advanced analytical and artificial intelligence scenarios. While the Proof of Concept focuses on analytical Data Products, the same governed mechanisms could be applied to cross-organizational machine learning pipelines, federated learning, or collaborative AI model development. These scenarios introduce new challenges related to governance, explainability, and responsibility, which merit further investigation.

Finally, the PoC reflects only a minimal instantiation of the proposed approach, involving a limited number of participants and Data Products. Although sufficient to demonstrate inter-organizational data sharing and governed Data Product derivation, this limited scale does not capture the full complexity and dynamics of larger Data Space ecosystems.

Applying and evaluating the proposed approach in more heterogeneous and realistic environments, involving multiple organizations across sectors and regulatory contexts, would provide valuable empirical insights and contribute to the maturation of Data Spaces as a practical paradigm for inter-organizational data collaboration.

REFERENCES

AWS. (2024). *What is a Data Lake? - Introduction to Data Lakes and Analytics - AWS*. Amazon Web Services, Inc. https://aws.amazon.com/what-is/data-lake

Armbrust, M., Das, T., Sun, L., Yavuz, B., Zhu, S., Murthy, M., Torres, J., van Hovell, H., Ionescu, A., Łuszczak, A., Świtakowski, M., Szafrański, M., Li, X., Ueshin, T., Mokhtar, M., Boncz, P., Ghodsi, A., Paranjpye, S., Senster, P., … Zaharia, M. (2020). Delta Lake: High-Performance ACID Table Storage over Cloud Object Stores. *Proceedings of the VLDB Endowment*, *13*(12), 3411–3424. https://doi.org/10.14778/3415478.3415560

Armbrust, M., Ghodsi, A., Xin, R., Zaharia, M., & Berkeley, U. (2021). *Lakehouse: A New Generation of Open Platforms that Unify Data Warehousing and Advanced Analytics.* In Proceedings of the Conference on Innovative Data Systems Research (CIDR).

Barr, M. (2020). *What is a Data Mesh - and How Not to Mesh it Up*. https://towardsdatascience.com/what-is-a-data-mesh-and-hownot- to-mesh-it-up-210710bb41e0

Costa, C., Andrade, C., & Santos, M. Y. (2019). Big Data Warehouses for Smart Industries. In *Encyclopedia of Big Data Technologies* (pp. 341–351). Springer International Publishing. https://doi.org/10.1007/978-3-319-77525-8_204

Costa, C., & Santos, M. Y. (2017). Big Data: State-of-the-art concepts, techniques, technologies, modeling approaches and research challenges. *IAENG International Journal of Computer*

Curry, E., & Tuikka, T. (2022). An Organizational Maturity Model forData Spaces: A Data Sharing Wheel Approach. In *Data Spaces* (pp. 21–42). Springer International Publishing. https://doi.org/10.1007/978-3-030-98636-0_2

Databricks. (2023). *Introduction to Data Lakes.* Retrieved February 10, 2025, from https://www.databricks.com/discover/data-lakes

Davenport, T. (2006). Competing on Analytics. *Harvard Business Review*.

Dehghani, Z. (2019). *How to Move Beyond a Monolithic Data Lake to a Distributed Data Mesh*. https://martinfowler.com/articles/data-monolithto- mesh.html

European Commission. (2023). What is a Data Space? - Common European Data Spaces. *European Strategy for Data*. https://ec.europa.eu/newsroom/lds/items/788753/en

Franklin, M., Halevy, A., & Maier, D. (2005). From databases to dataspaces. *ACM SIGMOD Record*, *34*(4), 27–33. https://doi.org/10.1145/1107499.1107502

Gandomi, A., & Haider, M. (2015). Beyond the hype: Big data concepts, methods, and analytics. *International Journal of Information Management*, *35*(2), 137–144. https://doi.org/10.1016/j.ijinfomgt.2014.10.007

Gelhaar, J., & Otto, B. (2020). Challenges in the Emergence of Data Ecosystems. *PACIS*.

Golfarelli, M., & Rizzi, S. (2009). Data warehouse design: Modern principles and methodologies. *McGraw-Hill, Inc.*

Grossman, R. L., & Siegel, K. P. (2014). Organizational Models for Big Data and Analytics. *Journal of Organization Design*.

Halevy, A., Franklin, M., & Maier, D. (2006). Principles of dataspace systems. *Proceedings of the Twenty-Fifth ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*, 1–9. https://doi.org/10.1145/1142351.1142352

Ribeiro, V., Barata, J., & Rupino Da Cunha, P. (2024). *A Maturity Model for Data Governance in Decentralized Business Operations: Architecture and Assessment Archetypes*.

Inmon, W. H. (2005). *Building the Data Warehouse, Fourth Edition*.

International Data Spaces Association. (2019). *IDS Reference Architecture Model (Version 3.0). https://internationaldataspaces.org*

International Data Spaces Association. (2024). *Making the dataspace protocol an international standard* (Version 1.0, July 2024). *https://internationaldataspaces.org*

Islam, S., & Apu, K. U. (2024). DECENTRALIZED VS. CENTRALIZED DATABASE SOLUTIONS IN BLOCKCHAIN: ADVANTAGES, CHALLENGES, AND USE CASES. *Global Mainstream Journal of Innovation, Engineering & Emerging Technology*, *3*(4), 58–68. https://doi.org/10.62304/jieet.v3i04.195

Khine, P. P., & Wang, Z. S. (2018). Data lake: a new ideology in big data era. *ITM Web of Conferences*, *17*, 03025. https://doi.org/10.1051/itmconf/20181703025

Kimball, R., & Ross, M. (2013). *The data warehouse toolkit: The definitive guide to dimensional modeling*. John Wiley & Sons.

Krishnan, K. (2013). *Data Warehousing in the Age of Big Data*. Morgan Kaufmann.

Laney, D. (2001). *3D Data Management: Controlling Data Volume, Velocity, and Variety.* META Group Research Note.

Larman, C. (2004). Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development. Analysis

Machado, I. A., Costa, C., & Santos, M. Y. (2022a). Data Mesh: Concepts and Principles of a Paradigm Shift in Data Architectures. *Procedia Computer Science*, *196*, 263–271. https://doi.org/10.1016/j.procs.2021.12.013

Machado, I. A., Costa, C., & Santos, M. Y. (2022b). *Proposal of an approach for the design and implementation of a data mesh*. https://hdl.handle.net/1822/82290

Manyika, J. (2011). Big data: The next frontier for innovation, competition, and productivity. *McKinsey Global Institute*.

Max, S., & Arif, W. (2020). *Data mesh in practice: how Europe's leading online platform for fashion goes beyond the data lake*. ThoughtWorks. https://www.slideshare.net/slideshow/data-mesh-in-practice-how-europes-leading-online-platform-for-fashion-goes-beyond-the-data-lake/235491516

Miloslavskaya, N., & Tolstoy, A. (2016). Big Data, Fast Data and Data Lake Concepts. *Procedia Computer Science*, *88*, 300–305. https://doi.org/10.1016/j.procs.2016.07.439

Nagel, L. , H., J. J., Perea, E., Lycklama, D., Mertens, C., Taillandier, A. S., & Grguric, A. (2022). *Design Principles for Data Spaces: Position Paper* (RWTH-2022-09082).

Otto, B. (2011). Organizing Data Governance: Findings from the Telecommunications Industry and Consequences for Large Service Providers. *Communications of the Association for Information Systems*, *29*. https://doi.org/10.17705/1CAIS.02903

Otto, B., Steinbuss, S., Teuscher, A., & Lohmann, S. (2019). Industrial Data Spaces Reference Architecture Model. *CERN European Organization for Nuclear Research*. https://doi.org/10.5281/zenodo.5105529

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A Design Science Research Methodology for Information Systems Research. *Journal of Management Information Systems*, *24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Regulation (EU) 2022/868 of the European Parliament and of the Council of 30 May 2022 on European data governance and amending Regulation (EU) 2018/1724 (Data Governance Act). (2022). *Official Journal of the European Union*, L 152, 1–44. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32022R0868

Ribeiro, V., Barata, J., & Cunha, P. R. D. (2024, August). A Maturity Model for Data Governance in Decentralized Business Operations: Architecture and Assessment Archetypes. In International Conference on Information Systems Development (ISD) (No. 32nd).

Rülicke, L., Fehrle, F., Martin, A., Monti, A., Berkhout, V., Warweg, O., & Möller, S. (2024). Exploring decentralized data management: a case study of changing energy suppliers in Germany. *Energy Informatics*, *7*(1), 8. https://doi.org/10.1186/s42162-024-00315-5

Russom, P. (2016). *Data Warehouse Modernization In the Age of Big Data Analytics BEST PRACTICES REPORT Q2 2016*.

ThoughtWorks. (2020). *Data Mesh. Retrieved 15 December 2020*, from https://www.thoughtworks.com/radar/techniques/data-mesh

Santos, M. Y., & Costa, C. (2020). Big data: Concepts, warehousing, and analytics. River Publishers.

Velu, C. K., Madnick, S. E., & van Alstyne, M. W. (2013). Centralizing Data Management with Considerations of Uncertainty and Information-Based Flexibility. *Journal of Management Information Systems*, *30*(3), 179–212. https://doi.org/10.2753/MIS0742-1222300307

APPENDIX

Appendix 1 – Participant Registry (participants.yml)

```yaml
version: "1.0"
last_updated: "2026-01-03"

participants:
  - participant_id: "org-a-customer-services"
    name: "Organization A (Customer Services)"
    type: "organization"
    domain: "crm"
    repo: "domain-crm"
    roles:
      - "provider"
    contacts:
      - role: "data_product_owner"
        name: "Customer Services Owner"
        email: "crm-owner@example.com"
    trust:
      status: "trusted"          # trusted | probation | suspended
      rationale: "PoC participant" #(trusted by default)
      valid_from: "2026-01-01"

  - participant_id: "org-b-ops-delivery"
    name: "Organization B (Operations & Delivery)"
    type: "organization"
    domain: "ops"
    repo: "domain-ops"
    roles:
      - "provider"
    contacts:
      - role: "data_product_owner"
        name: "Operations Owner"
        email: "ops-owner@example.com"
    trust:
      status: "trusted"
      rationale: "PoC participant" #(trusted by default)
      valid_from: "2026-01-01"

  - participant_id: "org-c-analytics-provider"
    name: "Organization C (Analytics Provider)"
    type: "organization"
    domain: "analytics"
    repo: "domain-analytics"
    roles:
      - "consumer"
      - "provider"
```

```
    contacts:
      - role: "data_product_owner"
        name: "Analytics Owner"
        email: "analytics-owner@example.com"
    trust:
      status: "trusted"
      rationale: "PoC participant" #(trusted by default)
      valid_from: "2026-01-01"
```

Figure 8: PoC Participants: *dataspace-registry\participants\participants.yml*

Appendix 2 – Data Product Specification (product.yml)

```
version: "1.0"

data_product:
  id: "crm.customer_profile.v1"
  name: "Customer Profile"
  domain: "crm"
  owner:
    participant_id: "org-a-customer-services"
    team: "Customer Services Data Team"
    contact: "crm-owner@example.com"
  description: >
    Domain-owned customer profile dataset providing stable customer
identifiers and
    contextual attributes (e.g., country, subscription type, status). This
Data Product is
    designed for cross-domain analytics while maintaining strict handling
rules for direct identifiers.
  lifecycle:
    status: "active"
    created_on: "2026-01-01"
    last_updated: "2026-01-03"
  contract:
    data_classification: "confidential"
    contains_pii: true
    pii_fields:
      - "full_name"
    allowed_purposes:
      - "analytics"
      - "reporting"
    prohibited_uses:
      - "individual_targeting"
      - "pii_export_or_redistribution"
    retention:
      max_retention_days: 90
    access_policy:
      default: "deny"
```

```
      grant_process: "github_issue_approval"
      required_fields:
        - "purpose"
        - "duration"
        - "requester_participant_id"
      obligations:
        - "Consumers must not export or store PII outside the governed
environment."
        - "Consumers should prefer non-PII interfaces when available."
  interface:
    type: "table_or_view"
    platform: "databricks"
    # Tip for PoC: implement this as a VIEW that excludes `full_name` for
cross-domain consumption.
    identifier: "main.crm_customer_profile_vw"
  schema:
    format: "delta"
    primary_key: ["customer_id"]
    columns:
      - name: "customer_id"
        type: "string"
        description: "Unique customer identifier within the Data Space context
(domain-owned key)."
      - name: "full_name"
        type: "string"
        description: "Customer full name (direct identifier; PII)."
      - name: "country_code"
        type: "string"
        description: "Customer country code (ISO 3166-1 alpha-2)."
      - name: "subscription_type"
        type: "string"
        description: "Subscription category or plan type."
      - name: "customer_status"
        type: "string"
        description: "Customer lifecycle status (e.g.,
active/suspended/churned)."

      - name: "created_at"
        type: "timestamp"
        description: "Timestamp when the record was created in the source
system."
      - name: "updated_at"
        type: "timestamp"
        description: "Timestamp of last update to the record."
      - name: "source_system"
        type: "string"
        description: "Identifier of the originating system."
      - name: "record_version"
        type: "integer"
```

```
        description: "Record version for change tracking / contract evolution
support."
      - name: "pii_flag"
        type: "boolean"
        description: "Indicates whether the record contains personally
identifiable information."
```

Figure 9: PoC Data Product Definition: *domain-crm\customer_profile\product.yml*

Appendix 3 – Baseline Data Contract (baseline_contract.yml)

```
version: "1.0"
contract_type: "baseline"
data_product_id: "crm.customer_profile.v1"
provider_participant_id: "org-a-customer-services"
contract_id: "crm.customer_profile.v1.baseline"

scope:
  interface:
    type: "table_or_view"
    identifier: "main.crm_customer_profile_vw"
  primary_key: ["customer_id"]
  contains_pii: true
  pii_fields:
    - "full_name"
usage_terms:
  allowed_purposes:
    - "analytics"
    - "reporting"
  prohibited_uses:
    - "individual_targeting"
    - "pii_export_or_redistribution"
  retention:
    max_retention_days: 90
  sharing:
    onward_sharing_allowed: false
  security:
    access_model: "deny_by_default"
    grant_required: true
    grant_process: "github_issue_approval"
service_levels:
  freshness:
    expected_update_frequency: "daily"
    maximum_data_delay_hours: 48
  availability:
    target: "best_effort"
  support:
    contact: "crm-owner@example.com"
data_quality_expectations:
```

```yaml
  rules:
    - id: "CP-001"
      description: "customer_id must be present"
      type: "not_null"
      column: "customer_id"
      severity: "error"
    - id: "CP-002"
      description: "country_code must follow ISO 3166-1 alpha-2 (2 uppercase
letters)"
      type: "regex"
      column: "country_code"
      pattern: "^[A-Z]{2}$"
      severity: "warning"
    - id: "CP-003"
      description: "customer_status must be one of: active, suspended,
churned"
      type: "accepted_values"
      column: "customer_status"
      values: ["active", "suspended", "churned"]
      severity: "warning"
    - id: "CP-004"
      description: "record_version must be >= 1"
      type: "min_value"
      column: "record_version"
      min: 1
      severity: "warning"
schema_policy:
  guarantees:
    - "No breaking schema changes within v1.* (additive changes allowed)."
    - "Primary key semantics remain stable (customer_id identifies a customer
record)."
  evolution:
    versioning_strategy: "semantic_versioning"
    breaking_change_requires_major_version: true
auditability:
  required_fields:
    - "updated_at"
    - "source_system"
    - "record_version"
  notes: "Consumers should use record_version and updated_at for change
tracking."
references:
  specification_ref: "domain-crm/customer_profile/product.yml"
```

Figure 10: PoC Data Contract: *customer_profile\contracts\baseline_contract.yml*

Appendix 4 – Interaction Data Contract (org-c-analytics-provider.yml)

```yaml
version: "1.0"
contract_type: "interaction"
data_product_id: "ops.operational_metrics.v1"
provider_participant_id: "org-b-ops-delivery"
consumer_participant_id: "org-c-analytics-provider"
contract_id: "ops.operational_metrics.v1.org-c-analytics-provider"

based_on:
  baseline_contract_ref: "domain-
ops/operational_metrics/contracts/baseline_contract.yml"
  baseline_version: "1.0"
scope:
  interface:
    type: "table_or_view"
    identifier: "main.ops_operational_metrics_vw"
  data_minimization:
    included_columns:
      - "customer_id"
      - "metric_date"
      - "tickets_opened"
      - "tickets_closed"
      - "avg_resolution_time_hours"
      - "sla_breaches"
      - "data_quality_score"
      - "updated_at"
      - "source_system"
usage_terms:
  purpose:
    description: "Join operational signals with customer identifiers to
compute engagement insights."
  allowed_purposes:
    - "analytics"
  retention:
    max_retention_days: 90
  onward_sharing_allowed: false
governance:
  approval_required: true
  approval_reference_type: "github_issue"
  enforcement:
    method: "platform_permissions"
    notes: "Access is granted via Databricks SELECT on the agreed interface."
data_quality_expectations:
  inherited: true
  additional_rules:
    - id: "OM-C-001"
      description: "sla_breaches must be >= 0"
      type: "min_value"
```

```
      column: "sla_breaches"
      min: 0
      severity: "warning"
schema_policy:
  inherited: true
  consumer_compatibility:
    notes: "Consumer expects daily granularity and depends on included_columns
only."
references:
  specification_ref: "domain-ops/operational_metrics/product.yml"
```

Figure 11: PoC Data Contract: *operational_metrics\contracts\org-c-analytics-provider.yml*

Appendix 5 – Access Grants Registry (access_grants.yml)

```
version: "1.0"
last_updated: "2026-01-03"

# Deny-by-default.
# Approved grants are recorded here. In this PoC, enforcement is procedural:
# access is considered valid only if a matching grant exists and references an
agreed interaction contract.

access_grants:
  - grant_id: "grant-0001"
    requester_participant_id: "org-c-analytics-provider"
    provider_participant_id: "org-a-customer-services"
    data_product_id: "crm.customer_profile.v1"
    contract_ref: "domain-crm/customer_profile/contracts/org-c-analytics-
provider.yml"
    contract_type: "interaction"
    contract_version: "1.0"
    access:
      permission: "read"
      interface: "table_or_view"
      scope:
        dataset: "customer_profile"
        restrictions:
          - "no_pii_export"
          - "analytics_only"
    purpose:
      description: "Compute engagement and churn indicators (input for
Engagement Insights)."
    validity:
      valid_from: "2026-01-03"
      valid_to: "2026-03-31"
    approval:
      approved_by: "crm-owner@example.com"
```

```
      approved_on: "2026-01-03"
      reference:
        type: "github_issue"
        id: "domain-crm#1"

  - grant_id: "grant-0002"
    requester_participant_id: "org-c-analytics-provider"
    provider_participant_id: "org-b-ops-delivery"
    data_product_id: "ops.operational_metrics.v1"
    contract_ref: "domain-ops/operational_metrics/contracts/org-c-analytics-
provider.yml"
    contract_type: "interaction"
    contract_version: "1.0"
    access:
      permission: "read"
      interface: "table_or_view"
      scope:
        dataset: "operational_metrics"
        restrictions:
          - "analytics_only"
    purpose:
      description: "Join operational signals with customer identifiers to
compute Engagement Insights."
    validity:
      valid_from: "2026-01-03"
      valid_to: "2026-03-31"
    approval:
      approved_by: "ops-owner@example.com"
      approved_on: "2026-01-03"
      reference:
        type: "github_issue"
        id: "domain-ops#1"
```

Figure 12: PoC Access Grants: *dataspace-registry\governance\access_grants.yml*

Appendix 6 – Federated Catalog (catalog.json)

```
{
  "version": "1.0",
  "last_updated": "2026-01-03",
  "data_products": [
    {
      "id": "crm.customer_profile.v1",
      "name": "Customer Profile",
      "domain": "crm",
      "owner_participant_id": "org-a-customer-services",
      "status": "active",
      "repo": "domain-crm",
```

```
      "paths": {
        "spec": "domain-crm/customer_profile/product.yml",
        "baseline_contract": "domain-
crm/customer_profile/contracts/baseline_contract.yml",
        "interaction_contracts": [
          "domain-crm/customer_profile/contracts/org-c-analytics-provider.yml"
        ]
      },
      "interface": {
        "type": "table_or_view",
        "platform": "databricks",
        "identifier": "main.crm_customer_profile_vw"
      },
      "tags": ["crm", "customer", "pii", "foundation"]
    },
    {
      "id": "ops.operational_metrics.v1",
      "name": "Operational Metrics",
      "domain": "ops",
      "owner_participant_id": "org-b-ops-delivery",
      "status": "active",
      "repo": "domain-ops",
      "paths": {
        "spec": "domain-ops/operational_metrics/product.yml",
        "baseline_contract": "domain-
ops/operational_metrics/contracts/baseline_contract.yml",
        "interaction_contracts": [
          "domain-ops/operational_metrics/contracts/org-c-analytics-
provider.yml"
        ]
      },
      "interface": {
        "type": "table_or_view",
        "platform": "databricks",
        "identifier": "main.ops_operational_metrics_vw"
      },
      "tags": ["ops", "support", "metrics", "dq"]
    },
    {
      "id": "analytics.engagement_insights.v1",
      "name": "Engagement Insights",
      "domain": "analytics",
      "owner_participant_id": "org-c-analytics-provider",
      "status": "active",
      "repo": "domain-analytics",
      "paths": {
        "spec": "domain-analytics/engagement_insights/product.yml",
        "baseline_contract": "domain-
analytics/engagement_insights/contracts/baseline_contract.yml",
```

```
      "interaction_contracts": []
    },
    "interface": {
      "type": "table_or_view",
      "platform": "databricks",
      "identifier": "main.analytics_engagement_insights_vw"
    },
    "tags": ["analytics", "insights", "derived", "lineage"]
  }
]}
```

Figure 13: PoC Federated Catalog: dataspace-registry\catalog\catalog.json